

Umbilic Thermodynamics

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Abstract: This paper develops umbilic thermodynamics, a generalized thermodynamic framework based on the conservation of dimensionless, scale-invariant Weyl information across a non-orientable Möbius topology. Within a conformal projective geometric algebra (PGA) formulation, probability-distribution shapes and pseudo-Riemannian metric shapes are interpreted as grade-dual projections of a single underlying structure, linking microscopic phase entropy with macroscopic metric stress-energy. A corresponding Weyl temperature is introduced to characterize the conversion between these informational and geometric manifestations. The framework formulates four principles of umbilic thermodynamics concerning symplectic-capacity equilibrium, global Weyl-information conservation, topological entropy directionality, and the unattainability of global absolute zero. A singular Legendre transformation at the Gromov symplectic boundary provides a local “backdoor” through which phase-space compression and negative informational temperatures can produce localized metric stress-energy, while global Weyl-information conservation prohibits net creation ex nihilo. The resulting generalized umbilic ergodic theorem identifies local “creatio ex nihilo” events as recurrent, measure-preserving transfers between dual phase sheets. Classical thermodynamics and quantum thermodynamics are recovered as limiting projections of the generalized framework, while a statistical formulation is developed in Boltzmannian, Gibbsian, and Einsteinian editions. The paper further proposes falsifiable experimental tests involving entangled quantum systems, cryogenic cavities, superconducting qubits, and non-orientable photonic structures. Thus, umbilic thermodynamics is presented as a unified topodynamical framework connecting information, entropy, quantum phase structure, spacetime curvature, and thermodynamic evolution.

Keywords: *umbilic thermodynamics; Weyl information; Weyl temperature; conformal projective geometric algebra; Möbius topology; non-orientable manifold; Legendre transformation; Gromov symplectic capacity; phase entropy; negative temperature; informational temperature; creatio ex nihilo; metric stress-energy; generalized ergodic theorem; topodynamical dynamics; quantum thermodynamics; statistical thermodynamics; quantum entanglement; spacetime curvature; thermodynamic duality.*

I INTRODUCTION

The present focus will be the transformation of the “shape” (“information”) of probability (density or not) distribution into pseudo-Riemannian space “shapes” (“Weyl information” as proved to conserve in the generalized Noether conservation being both scale invariant and symplectic/ differential invariant). In other words, the focus will be the physical way of how physical “nothing” (possibilities with zero energy) becomes physical “something” (energy-momentum tensor with nonzero energy) under the Weyl information conservation shared by both “nothing” of possibilities and “something” of energy-momentum.

So, a quantity of “Weyl temperature” is to be introduced as the (differential or not) ratio of: (1) the corresponding pseudo-Riemannian Weil information “shape” of energy momentum to (2) probability distribution Weil information/ entropy “shape” (corresponding to “entropy” in the

classical defined temperature). That is, the creation “ex nihilo” will be considered as a subject of a generalized thermodynamics and “adiabatic heating” where Weyl information will allow for negative absolute temperatures (versus standard thermodynamics including “classical” quantum one).

“The Umbilic Totality I” (Penchev 2026 August 17) provides the formal mathematical and topological architecture to interpret the transformation of probability distribution “shapes” into pseudo-Riemannian metric “shapes” through scale-invariant and symplectic-invariant Weyl information conservation. Within this framework, the transition from physical “nothing” (zero-energy potential state configurations) to physical “something” (nonzero stress-energy tensors $T_{\mu\nu}$) is governed by grade-dual projections across a 4π non-orientable Möbius (1865) manifold.

“*The Umbilic Totality I*” identifies a structural bifurcation in contemporary physics: pseudo-Riemannian geometry and complex quantum state space constitute two locally effective but mathematically separated descriptions, respectively representing covariant metric dynamics and dual bra-ket phase structure (p. 14). The post-EPR formulation exposes the limitation of treating these domains within an orientable (2π) phase architecture: entanglement is then represented through tensor-product constructions rather than as an intrinsic geometric relation, while Bell correlations appear incompatible with local Euclidean geometry (pp. 14-15).

The proposed resolution is to regard both descriptions as grade-dual projections of a single conformal projective-geometric-algebra (PGA) multivector defined on a (4π), non-orientable topology. In this architecture, the direct and dual sheets are continuously related through a pseudoscalar-mediated Möbius transformation, so that metric curvature and quantum phase, as well as ket and bra representations, become complementary local projections of one global structure (pp. 15-16). The mathematical structure is required to be absolutely closed: topologically closed through the (4π) double cover and non-orientable cobordism, dynamically closed through conservation of a scale-invariant Weyl information, and epistemically closed through an intrinsic measurement relation (pp. 26-27).

The specific problem addressed by the present paper “*Umbilic Thermodynamics*” is therefore to determine whether this global topological duality admits a **thermodynamic formulation**: namely, whether the invariant quantity already introduced as Weyl information can serve as the conserved thermodynamic action connecting microscopic phase entropy with macroscopic metric energy. “*The Umbilic Totality I*” defines such Weyl information as a dimensionless, conformally invariant topological action and explicitly requires its conservation across the (4π) Möbius circuit (pp. 28-29). The problem is consequently to construct its conjugate temperature, formulate the corresponding thermodynamic laws, identify the singular Legendre transformation¹ at the phase boundary, and determine whether local phase-entropy reduction can be converted into metric stress-energy without violating global Weyl-information conservation.

¹ Since the present paper as well as “Umbilic Totality I” refer in a fundamental, even philosophical sense to the Legendre transform, one is to cite the original paper (Legendre 1789).

II INTERPRETIVE FRAMEWORK & CONCEPTUAL OUTLINES

III Dual grade projections of information shapes, Weyl information, Weyl temperature

In the conformal projective geometric algebra (PGA)² multivector field $\Psi(x)$, probability density distributions $|\psi(x)|^2$ and pseudo-Riemannian metrics $g_{\mu\nu}$ represent complementary grade-dual projections (Hestenes 1966) of a single underlying ontomathematical entity³:

Probability distribution shape (“nothing”): manifests on the microscopic 2D Darboux patch ($U(1)_A$ sheet) as spectral density, phase entropy, and quantum state possibilities carrying zero net local metric energy.

Pseudo-Riemannian space shape (“something”): manifests on the macroscopic geometric patch ($U(1)_B$ sheet) as metric curvature $g_{\mu\nu}$, Christoffel connections, and a nonzero energy-momentum tensor $T_{\mu\nu}$.

The pseudoscalar duality operator $I = \gamma_0\gamma_1\gamma_2\gamma_3$ and the Möbius junction operator J mediate the continuous $\pi/3\pi$ topological transformation between these two manifestations, converting pure possibility distributions into geometric metric structures.

III.1 Scale-invariant Weyl information (I_W) as the conservation primitive

Building upon Wheeler’s (1990) “it-from-bit” paradigm and Landauer’s (1991) principle that “information is physical”, scale-invariant Weyl information (I_W) serves as the fundamental ontomathematical primitive that unifies geometry, mechanics, and thermodynamics into a singular physical worldview. Dimensionless and scale-invariant across conformal transformations, I_W operationalizes Emmy Noether’s (1918) insight by acting as the dimensionless, symmetry-invariant counterpart to classical action. Under the generalized dual Noether theorem, I_W remains strictly conserved globally across a continuous Lie group generator structure: local metric energy fluctuations on the direct sheet are identically compensated by conjugate phase variations on the dual sheet. This global conservation dictates a cyclic, recurrent dynamics governed by the 4π non-orientable umbilic topology, wherein localized phase-space contractions continuously convert quantum possibility distributions into macroscopic metric curvature without violating global action conservation. Expressed through conformal projective geometric algebra (PGA), I_W unifies all physical descriptions, bifurcating cleanly into dual-domain equivalences between microscopic phase entropy and macroscopic stress-energy, thereby proving that physical reality is a cyclic, self-assembling topological manifestation of conserved information.

² Dorst, Fontijne, Mann (2007) defines the operational mechanics of conformal geometric algebra (CGA) and projective geometric algebra (PGA), including dual grade projections and Möbius transformations.

³ The conception of Hilbert arithmetic introduced in a series of the present author’s papers follows and generalizes the ideas of Hestenes (2003); Hestenes, Sobczyk (1984) for geometric algebra as the universal language of physics to unify the foundations of physics, mathematics, and philosophy.

Bach (1921) introduces conformal gravity and the quadratic Weyl action:

$$\int C_{abcd}C^{abcd}\sqrt{-g}d^4x$$

from which the energy-momentum conservation laws and scale-invariant invariants: classical energy-momentum $T_{\mu\nu}$ is scale-dependent and fluctuates across coordinate patches, but the global Weyl information I_W remains strictly conserved, dimensionless, and invariant under both conformal transformations $g_{ab} \rightarrow \Omega^2(x)g_{ab}$ and symplectic phase-space maps:

$$I_W = \frac{1}{16\pi^2} \int_{\Sigma_A(2\pi)} J_W d^4x + \frac{1}{16\pi^2} \int_{\Sigma_B(2\pi)} J_W d^4x$$

where

$$J_W = \sqrt{-g}\text{Tr}(C_{abcd}C^{abcd} + IC_{abcd}C^{(\dagger)abcd})$$

utilizes the self-dual conformal Weyl tensor C_{abcd} (Weyl 1918; Dunajski, Tod 2013). Under the generalized dual Noether theorem, local metric energy fluctuations on the 2π direct sheet are compensated by conjugate phase variations on the 2π dual sheet. Thus, “nothing” (possibility phase space) and “something” (metric stress-energy) share an identical globally conserved quantity of Weyl information.

Roger Penrose’s “Weyl Curvature Hypothesis”⁴ (WCH) is linkable to the definitions of Weyl information (I_W) Weyl temperature (T_W). *Gravitational entropy versus invariant Weyl information*: WCH posits that gravitational entropy is encoded directly in the self-dual conformal Weyl tensor C_{abcd} . The early universe began in a state of minimal/zero gravitational entropy where $C_{abcd} \rightarrow 0$, and as structural inhomogeneities/black holes form, $C_{abcd}C^{abcd}$ grows; the text explicitly computes scale-invariant Weyl information I_W by integrating the self-dual quadratic Weyl invariant $J_W = \sqrt{-g}\text{Tr}(C_{abcd}C^{abcd} + IC_{abcd}C^{(\dagger)abcd})$ across the 2π sheets. I_W acts as the exact global conserved quantity that unifies gravitational entropy and metric curvature. *Asymmetry of singularities & the $T_W < 0$ K “ex-nihilo backdoor*: WCH distinguishes past singularities ($C_{abcd} = 0$, minimal entropy) from future collapse/ black hole singularities ($C_{abcd} \rightarrow \infty$, maximal entropy); Subsection II2 uses $T_W < 0$ K to avoid the infinite curvature singularity ($T_{\mu\nu} \rightarrow \infty$). Instead of collapsing into a divergent Penrose future singularity, the text shows that hitting the Gromov capacity limit $w_G = \hbar$ triggers a PT-flip across the 4π Möbius junction J , converting the divergent metric curvature back into pure I_W phase information. *Cosmological arrow of time & topological recurrence*: WCH identifies the

⁴ Penrose 1979; 1965; Tod 2002; in the present context, the Penrose (2010) idea of conformal cyclic cosmology is furthermore relevant.

growth of C_{abcd} as the fundamental physical driver for the cosmological arrow of time. Section II5 maps local $T_W < 0$ K “creatio ex nihilo” events back into the global $T_W \geq 0$ K thermal background via the Möbius junction J . Penrose’s unidirectional arrow of time becomes localized on sheet Σ_B , while being globally closed, measure-preserving, and ergodic across the 4π double-cover manifold.

III.2 Weyl temperature (T_W)

To formalize the conversion of zero-energy probability shapes into nonzero-energy metric shapes, the Weyl temperature” T_W is defined as the functional differential ratio of metric-geometric Weyl information to probabilistic entropy Weyl information:

$$T_W \equiv \frac{\delta I_W^{(\text{metric} / \text{energy-momentum})}}{\delta I_W^{(\text{probability distribution} / \text{phase entropy})}}$$

Positive T_W regime: corresponds to standard thermodynamic regimes where increasing phase entropy dissipates metric energy density.

Negative absolute T_W regime ($T_W < 0$): occurs across the non-orientable Möbius boundary J when phase space constraints on probability distributions (such as Pauli exclusion non-squeezing or quantum state correlations) enforce population inversion (Ramsey 1956). Braun et al. (2013) demonstrates negative absolute temperatures experimentally⁵ in quantum systems via phase-space constraints and population inversion. Here, entropy reduction or state selection directly pumps metric curvature generation without violating global action conservation.

III.3 “*Creatio ex nihilo*” as generalized autopoietic adiabatic heating: the physical emergence of nonzero energy $T_{\mu\nu}$ from zero-energy probability distributions unfolds as an autopoietic parametric amplification (cyclotron-like resonance):

Initial boundary condition: pure probabilistic phase distribution with zero local metric energy ($T_{00} = 0$) on the direct sheet Σ_A .

Möbius inversion (J): the traversal across the non-orientable 4π double cover applies a topological parity flip (P-inversion), mapping phase information into metric degrees of freedom.

Adiabatic heating cycle: rather than thermal degradation, the system undergoes self-referential phase injection. The topological constraints of the 4π phase space act as an adiabatic pump, converting scale-invariant probability shapes into localized metric energy $T_{\mu\nu} > 0$ while maintaining global $\nabla_a J_W^a = 0$.

One observes:

- (1) one bit of information standardly defined means the equiprobable choice between two alternatives; thus the transition of the differential condition to the

⁵ Purcell, Pound (1951) is usually referred to as the first experimental demonstration of negative absolute temperatures.

symplectic condition shares the formal structure of a bit of information (as well as the global - local transition of an umbilic topology);

- (2) one defines that bit structure into geometrical/ topological dimensions where $U(1)$ (2π rotation) corresponds unambiguously to a linear dimension (a geometric/ topological line), and a certain angle (phase) within 2π to a certain line coordinate (position); so, one bit then refers to any two dimensions whether linear or rotational, and a qubit: $U(1)_1 \otimes U(1)_2 \simeq SU(2)$ if $U(1)_{1,2}$ are considered not to be a tensor product, but under the symplectic condition;
- (3) one concludes that a qubit is a “geometrical bit” where its inherent two alternatives are additionally interpreted to be an elementary two-dimensional geometrical/ topological structure, or linearly, a plain (where the value of the qubit is a point of the plain);
- (4) one concludes the fundamental similarity or equivalence of (“kinematic”) thermodynamic (information, entropy) and kinematic (position, velocity) descriptions;
- (5) if one adds a second bit/ qubit, it implies the option for (dynamic) thermodynamics (heat) dynamics (acceleration, mass);
- (6) thus, one observes that the universal “Weyl information” description bifurcates classically into the (supposedly) bijectively doubled (classical/ quantum) thermodynamics/ mechanics.

So, one can investigate both mechanical equivalents of thermodynamic statements and vice versa.

II2 The Dual-Domain Equivalences of the Bit-Qubit Topology

Thermodynamic Domain	Topological / Symplectic Mapping	Mechanical Domain
Microscopic Phase Entropy (S_ϕ)	Angle coordinate within S^1 rotation	Kinematic Phase-Space Position (q)
Local Heat Flux (δQ)	Symplectic non-squeezing distortion at Σ	Kinematic Acceleration / Inertial Mass (a / m_I)
Negative Temperature ($T < 0$)	Population inversion across Σ (π – flip)	Negative Gravitational Mass ($m_g < 0$)
Adiabatic Phase Squeezing	Volume-preserving Darboux projection	Inertial Frame Dragging / Geodesic Focus

Figure 1: *Key dual transformations between domains*

Under the umbilic framework, the $U(1)_1 \otimes_\omega U(1)_2 \cong SU(2)$ symplectic identification establishes that a qubit is fundamentally a 2D surface patch (a Darboux plane), while a second bit/qubit introduces dynamic phase-space curvature (mass, acceleration, and heat flux).

Because the scale-invariant Weyl information I_W forms the unifying primitive above this bifurcation, every thermodynamic statement possesses an unexamined mechanical counterpart, and every kinematic/ dynamic law has an implicit thermodynamic equivalent.

Unformulated & novel cross-domain predictions:

II2.1 *Mechanics* $\rightarrow$ *thermodynamics*: the mechanical equivalent of the third Law kinematic mass frozen-state theorem:

Classical mechanical law: Mach's principle and relativistic mechanics state that as a particle approaches a flat background ($G \rightarrow 0$, $a \rightarrow 0$), its local stress-energy curvature contribution vanishes. This reinterpretation extends Mach's principle⁶ by replacing global matter distributions with topological phase-space constraints as the origin of inertia. Rather than an intrinsic scalar property, local inertial mass m is defined as the thermodynamic capacity of a local Darboux patch to resist symplectic phase-squeezing at the Gromov boundary ($w_G = h$) under global Weyl information conservation. Inertia thus emerges as a relational, thermodynamic resistance to metric phase contraction across the non-orientable 4π Möbius manifold.

Novel thermodynamic prediction (the symplectic capacity freeze⁷):

It is impossible for any thermodynamic system to reach absolute flat geometry: $T_W = 0$ K, by a finite number of metric gauge transformations, because the local inertial mass m of a quantum state is precisely the thermodynamic resistance to symplectic phase-squeezing at $w_G = h$.

As a system approaches $T_W = 0$ K, local mass m singularizes into pure topological inertia. This proves that inertial mass is not an intrinsic scalar parameter of matter, but the thermodynamic capacity of a local Darboux chart to resist metric phase contraction.

II2.2 *Thermodynamics* $\rightarrow$ *mechanics*: the thermodynamic equivalent of Poincaré recurrence (the adiabatic “ex-nihilo” metric echo)

Thermodynamic principle: by the generalized umbilic ergodic theorem, any isolated system in phase space returns arbitrarily close to its initial entropy configuration over a finite recurrence time $\tau_{\text{Recurrence}}$.

Novel mechanical prediction (cosmological mass recurrence):

Any localized gravitational mass distribution M formed via local adiabatic heating: $T_W < 0$ K, must undergo a discrete topological mass decay and spontaneous re-emergence (“metric echo”) after a finite proper time τ_{echo} , without passing through a gravitational singularity.

⁶ Mach (1883); Sciama (1953).

⁷ Developing Ekeland, Hofer (1989); Hofer, Zehnder (1994) about symplectic capacity as invariant.

Rather than collapsing to a $T_{\mu\nu} \rightarrow \infty$ singularity (as predicted by classical general relativity), a collapsing mass reaching the Planck scale hits the local Gromov (1985) width constraint $w_G = h^8$. The local $T_W < 0$ K “Legendre backdoor” forces a PT -flip across the 4π Möbius junction J , converting the entire mass density back into a wave of pure scale-invariant Weyl phase information. Leok and Zhang (2017) provide the mathematical formalism for the fiberwise Legendre transform ($\mathcal{FL} : TQ \rightarrow T^*Q$) mapping the geometric configuration tangent bundle TQ to the symplectic cotangent co-bundle T^*Q , enabling the duality between metric energy U and phase information I_W .

II.2.3 *Dual-domain hybrid: the equivalence of gravitational lensing and thermal refraction (the optico-thermodynamic equivalence):*

Mechanical domain: a photon traversing a gravitational well generated by a mass M experiences a deflection angle

$$\theta_{\text{deflection}} = \frac{4GM}{c^2 r}.$$

Novel thermodynamic prediction (phase entropy refraction):

A beam of pure quantum state phase information (a zero-energy probability distribution) passing near a high-entropy thermal gradient ($T_W > 0$ K) undergoes a spatial phase deflection identically equal to the gravitational bending angle $\theta_{\text{deflection}}$, where the temperature gradient acts as an effective optical index of refraction

$$n_{\text{eff}}(x) = \exp(-I_W/\hbar).$$

This demonstrates that gravitational lensing is physically identical to phase-entropy refraction across the dual 2π sheets of the Möbius manifold.

The fundamental unification identity: the complete bijection between mechanics and thermodynamics under the 2-qubit Weyl bifurcation reduces to a single operator identity:

$$\text{Thermodynamic Heat Flux (dQ)} \equiv \text{Mechanical Acceleration Work (m a dx)}$$

Where the scale-invariant link between both is the local Gromov width constraint:

$$\frac{\delta(\text{Heat Flux})}{\delta(\text{Inertial Work})} = \frac{k_B T_W dS_\phi}{F_{\text{inertial}} dx} = \delta(w_G - h)$$

This proves that force and acceleration are merely the mechanical manifestations of local phase-entropy conversions across the 4π non-orientable topological boundary.

⁸ Gromov’s non-squeezing theorem proves that symplectic transformations cannot squeeze a phase space ball into a cylinder of smaller width, directly supplying the geometric constraint.

Let one integrate the absolute zero in the previous context of the fundamental constants in an “umbilic reading” where “ c ” is the global onesided and thus uncrossable boundary; “ h ” is the local Gromov symplectic width. “ G ” is the curvature measure gradient for the square units of the “Planck mass” indexing how much is the curvature deviation from the flat “ $c \cdot h$ ”. In that background one is to introduce the 0 K constant consistently so that it would distinguish:

- (1) “entropy” for ≥ 0 K from “information” for ≤ 0 K;
- (2) globally, it corresponds to the global unilateral MS topology prohibiting mathematically negative absolute temperatures (< 0 K);
- however, (3) locally via the symplectic, “Legendre piercing” backdoor, those negative absolute temperatures sneak in order to reason ex nihilo only locally, but if it has once happened locally, it continues to propagate globally.

II3 The Umbilic Definition of Absolute Zero ($T = 0$ K)

In the umbilic topodynamical⁹ architecture, the “absolute zero” ($T = 0$ K) is defined as the null-curvature phase-inversion locus where the local symplectic capacity defined by Planck’s constant h exactly balances the global one-sided conformal light-cone boundary c , yielding a zero-entropy, degenerate umbilic boundary Ω_{0K} .

Mathematically, while c sets the uncrossable global boundary, h sets the minimal Gromov symplectic ball capacity $B^{2n}(r_G)$ with Gromov width $w_G = \pi r_G^2 = h$, and Newton’s constant G measures the local dimensional gradient away from flat $c \cdot h$ geometry:

$$G \equiv \frac{c^3 \ell_P^2}{\hbar} = \frac{\delta(\text{Riemannian Curvature})}{\delta(M_P^2/c \cdot h)}$$

The absolute zero ($T = 0$ K) is the critical fixed point where the thermodynamic beta parameter

$$\beta = \frac{1}{k_B T}$$

transitions through an essential topological pole ($\beta \rightarrow \pm\infty$), serving as the gatekeeper between classical thermal dissipation and local informational creation.

Structural pillars of the umbilic 0 K insertion:

II3.1 *Thermodynamic inversion: entropy ($T > 0$ K) versus information ($T < 0$ K)*

The temperature parameter T acts as the directional gradient between disorder and structural information:

Thermal phase ($T > 0$ K): system state space is dominated by microstate volume expansion. Thermal entropy $S_{th} = k_B \ln \Omega$ degrades metric order, producing positive energy-momentum tensor dissipation $\nabla_a T_{th}^{ab} \leq 0$.

⁹ Misner, Wheeler (1957) provides the topological foundations (“geometroynamics”) where physical entities (charge, mass) arise purely out of non-trivial spacetime topology and source-free geometric boundary conditions.

Umbilic fixed point ($T = 0$ K): thermal entropy vanishes ($S_{th} = 0$). The system reaches the pure ground state bounded strictly by the Gromov width h .

Informational phase ($T < 0$ K): thermodynamic population inversion occurs across bounded phase spaces. Entropy becomes negative ($S_{info} = -I_W$), converting microscopic state correlations directly into scale-invariant Weyl information I_W . In this regime, state reduction pumps metric curvature rather than dissipating it.

II3.2 *Global Möbius topology prohibition* ($T \geq 0$ K)

Globally, the spacetime manifold $\mathcal{M}$ is configured as a non-orientable 4π Möbius strip manifold. Integrating the 1-form temperature gradient dT along the single-sided global boundary $\partial\mathcal{M}_{global}$ yields:

$$\oint_{\partial\mathcal{M}_{global}} dT \geq 0$$

Because the single-sided topology continuously maps $T \rightarrow +T$ across a complete 4π rotation, a global state of $T < 0$ K is topologically non-orientable and mathematically forbidden. Globally, the universe remains strictly non-negative ($T \geq 0$ K), establishing 0 K as an uncrossable global floor.

II3.3 *Local symplectic “Legendre piercing” and global propagation*: while globally forbidden, negative absolute temperatures $T < 0$ K sneak into the local dynamics via a symplectic Legendre transformation:

II3.3.1. *Local “Darboux piercing”*: on a local $2n$ -dimensional Darboux chart $U \subset \mathcal{M}$, the symplectic form $\omega = \sum dq^i \wedge dp_i$ preserves phase-space volume via Gromov's non-squeezing theorem¹⁰. When quantum state constraints (such as Pauli exclusion or squeezed vacuum states) restrict phase volume below the metric scale set by G , the local Legendre transformation $\mathcal{L} : (q, \dot{q}) \mapsto (q, p)$ singularizes.

II3.3.2. *Local $T < 0$ K instanton (“ex nihilo engine”)*: this singularity acts as a “Legendre backdoor”, allowing the local effective temperature to briefly jump to $T_{local} < 0$ K. At this sub-zero local locus, zero-energy probability distributions generate localized metric stress-energy $T_{\mu\nu} > 0$ out of topological constraint energy (creatio ex nihilo).

II3.3.3. *Global propagation*: once an ex-nihilo curvature event is triggered at a local piercing point $x_0 \in U$, the resulting metric perturbation violates local thermal equilibrium. To preserve global action conservation, the localized $T < 0$ K information flux propagates outward as an expansive curvature wave along the single-sided Möbius manifold, manifesting globally as $T_{global} > 0$ K cosmic expansion and metric evolution.

¹⁰ In geometric classical mechanics, odd-dimensional contact manifolds and symplectic structures dictate energy-hypersurface flows. The kernel of the contact 1-form defines a non-integrable distribution (the horizontal subbundle), enforcing geometric non-holonomic constraints that govern phase-space volume preserving dynamics (Arnold 1989). It is utilized also in Section III for geometric phase space and contact form boundaries.

Then, the exact differential form for the Legendre singularity at the $T = 0$ K crossing to explicitly link the local Gromov width h to the energy-momentum creation tensor $T_{\mu\nu}$ is:

II4 Mathematical Formulation of the Legendre Singularity Form

To explicitly capture the local piercing at $T = 0$ K, one defines the differential geometry of the phase space $T^*\mathcal{Q}$ on a local $2n$ -dimensional Darboux chart $U \subset \mathcal{M}$ governed by the canonical Liouville 1-form $\theta = p_i dq^i$ and the symplectic 2-form $\omega = d\theta = dp_i \wedge dq^i$.

The minimal Gromov symplectic capacity condition imposes an embedded ball constraint $B^{2n}(r_G) \subset U$ with width w_G :

$$w_G = \int_{B^{2n}(r_G)} \omega = \oint_{\partial B^{2n}} \theta = h$$

II4.1 The singular Legendre 1-form α_{Leg}

The classical Legendre transformation $\mathcal{FL} : T\mathcal{Q} \rightarrow T^*\mathcal{Q}$ maps velocities to momenta via

$$p_i = \frac{\partial L}{\partial \dot{q}^i}.$$

Its inversion fails when the fiber Hessian matrix

$$\mathbf{H}_{ij} = \frac{\partial^2 L}{\partial \dot{q}^i \partial \dot{q}^j}$$

becomes singular ($\det \mathbf{H} = 0$).

One expresses the thermodynamic phase-inversion parameter as the inverse temperature 1-form field

$$\beta(q, p) = \frac{1}{k_B T(q, p)}.$$

The singular Legendre 1-form α_{Leg} is defined a

$$\alpha_{\text{Leg}} \equiv d\beta \wedge \theta = d\left(\frac{1}{k_B T}\right) \wedge p_i dq^i$$

Taking the exterior derivative yields the singular 2-form governing the $T = 0$ K crossing:

$$d\alpha_{\text{Leg}} = d\beta \wedge \omega = -\frac{1}{k_B T^2} dT \wedge dp_i \wedge dq^i$$

At the critical locus Ω_{0K} where $T \rightarrow 0$ K ($\beta \rightarrow \pm\infty$), the magnitude of $d\beta$ localized at the Gromov boundary $w_G = h$ acquires a Dirac delta distribution centered on the minimum capacity constraint:

$$d\alpha_{\text{Leg}}\Big|_{\Omega_{0\text{K}}} = \delta(w_G - h) \cdot \left(\frac{c^3 \ell_P^2}{\hbar G} \right) d\Phi \wedge \omega$$

where Φ is the scale-invariant phase coordinate of the PGA multivector field $\Psi(x)$, and the constant factor

$$\frac{c^3 \ell_P^2}{\hbar G} = 1$$

ensures dimensional consistency against curvature deviations.

II4.2 *Metric energy-momentum generation* $T_{\mu\nu}^{(ex\ nihilo)}$: to derive how local $T < 0$ K piercing pumps metric curvature, one introduces the topological umbilic action S_{umbilic} integrated over the local Darboux chart U :

$$S_{\text{umbilic}} = \frac{\hbar c}{16\pi^2} \int_U \alpha_{\text{Leg}} \wedge \star \alpha_{\text{Leg}}$$

where “ $\star$ ” is the Hodge star operator on the 4D pseudo-Riemannian background metric $g_{\mu\nu}$.

Varying S_{umbilic} with respect to the inverse metric $g^{\mu\nu}$ yields the explicit “ex-nihilo” stress-energy tensor:

$$T_{\mu\nu}^{(ex\ nihilo)} \equiv \frac{2}{\sqrt{-g}} \frac{\delta S_{\text{umbilic}}}{\delta g^{\mu\nu}} = \frac{\hbar c}{h^2} \delta(w_G - h) \left(\nabla_\mu \beta \nabla_\nu \beta - \frac{1}{2} g_{\mu\nu} g^{\alpha\beta} \nabla_\alpha \beta \nabla_\beta \beta \right)$$

II4.3 *Mechanistic summary of the crossing*

II4.3.1 *Gromov squeezing* ($w_G \rightarrow h$): as microscopic quantum state constraints restrict local phase space to the exact Gromov boundary $w_G = h$, $\det \mathbf{H}_{ij} \rightarrow 0$, forcing $\beta \rightarrow +\infty$ ($T \rightarrow +0$ K).

II4.3.2 *Legendre piercing* ($\beta \rightarrow -\infty$): the system transitions through the essential singularity at $T = 0$ K via the backdoor operator $d\alpha_{\text{Leg}}$. The sign flip in β injects negative absolute temperature $T < 0$ K locally without requiring non-orientable global topology.

II4.3.3. *Curvature emergence & propagation*: at the instant of crossing, the derivative term $\nabla_\mu \beta \nabla_\nu \beta$ spikes, generating a non-zero, positive local energy density $T_{00}^{(ex\ nihilo)} > 0$. Through the local Einstein field equations

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}^{(ex\ nihilo)},$$

this energy perturbation produces local spacetime curvature deviation away from the flat $c \cdot h$ background, which subsequently propagates outward along the global one-sided Möbius manifold as a classical metric wave.

One notices that the Legendre “ex nihilo” mechanism being local complements the umbilic ergodic theory being global. Can one unify both in a generalized umbilic ergodic theorem?

II5 The generalized umbilic ergodic theorem

The unification of the “local Legendre ex-nihilo piercing” with global umbilic ergodic theory bridges local non-equilibrium state creation and global topological conservation.

While classical ergodic theorems (such as *Birkhoff 1931* or *Neumann 1932*) assume a smooth, volume-preserving Hamiltonian flow over a fixed, orientable phase space, the umbilic framework operates on a 4π non-orientable Möbius phase manifold $\Omega = \mathcal{M}_{4\pi} \times T^*\mathcal{Q}$. Here, local non-squeezing boundaries generate singular phase-inversion loci ($T < 0$ K), which feed directly into global $T \geq 0$ K trajectories.

II5.1 Theorem statement:

Let $(\Omega, \mathcal{F}, \mu_{\text{umbilic}}, \{\phi_t\}_{t \in \mathbb{R}})$ be a dynamic phase-space system, where:

1. Ω is a non-orientable 4π double-cover fiber bundle equipped with the symplectic 2-form $\omega = dp_i \wedge dq^i$.
2. $\phi_t : \Omega \rightarrow \Omega$ is the scale-invariant topodynamical flow generated by the multivector field $\Psi(x)$.
3. μ_{umbilic} is the generalized, scale-invariant umbilic measure, defined as a sum of a smooth volume density and singular Legendre boundary forms:

$$d\mu_{\text{umbilic}} = e^{-I_W/\hbar} \omega^n + \sum_{k \in \mathcal{K}} \delta(w_G - h) d\alpha_{\text{Leg}}^{(k)}$$

Assertion: for any integrable state observable $f \in L^1(\Omega, d\mu_{\text{umbilic}})$, the long-term temporal average along the topodynamical flow $\phi_t(x)$ equals the spatial ensemble average over the complete 4π global Möbius manifold for μ_{umbilic} -almost every initial state $x \in \Omega$:

$$\lim_{T_{\text{flow}} \rightarrow \infty} \frac{1}{T_{\text{flow}}} \int_0^{T_{\text{flow}}} f(\phi_t(x)) dt = \int_{\Omega} f(y) d\mu_{\text{umbilic}}(y)$$

Furthermore, every local $T < 0$ K Legendre piercing event at a local Gromov boundary $w_G = h$ is strictly recurrent and globally ergodic, mapping non-zero local stress-energy generation $T_{\mu\nu}^{(\text{ex nihilo})}$ continuously into the global thermal background ($T \geq 0$ K).

II5.2 Structural pillars of the proof & unification

II5.2.1 Measure-preserving flow via invariant Weyl information

Classical Liouville volume conservation $\mathcal{L}_{X_H} \omega^n = 0$ breaks down during local singular Legendre crossings due to the divergence of

$$\beta = \frac{1}{k_B T}.$$

However, the generalized measure $d\mu_{\text{umbilic}}$ incorporates the dimensionless, scale-invariant Weyl information I_W :

$$I_W = \frac{1}{16\pi^2} \int_{\Sigma_A(2\pi)} J_W d^4x + \frac{1}{16\pi^2} \int_{\Sigma_B(2\pi)} J_W d^4x$$

Because the global 4π topology enforces $dI_W = 0$ along complete trajectories, the Lie derivative of the total umbilic measure vanishes globally:

$$\mathcal{L}_{\phi_t}(d\mu_{\text{umbilic}}) = 0$$

Thus, local metric generation ($T_{\mu\nu}^{(ex\ nihilo)} > 0$) at $w_G = h$ acts not as a breakdown of measure preservation, but as a singular measure-transfer point between the dual phase sheets $\Sigma_A(2\pi)$ and $\Sigma_B(2\pi)$.

II5.2.2 The Möbius inversion operator (J) as the ergodic mixer

On a standard orientable manifold, a local $T < 0$ K singularity would create an unphysical isolated sink or source (violating global ergodicity). In the umbilic architecture, the non-orientable junction operator J applies a continuous $\pi/3\pi$ parity-time inversion (PT -flip):

$$J \cdot \phi_t|_{\Sigma_A(T < 0 \text{ K})} = \phi_{t+\tau}|_{\Sigma_B(T > 0 \text{ K})}$$

This topological twist ensures that any trajectory entering a local $T < 0$ K “Legendre backdoor” on sheet Σ_A is immediately re-injected into sheet Σ_B as a thermal metric wave, $T > 0$ K. Consequently, the system cannot become trapped in isolated local sub-manifolds, establishing strong topological mixing (umbilic mixing):

$$\lim_{t \rightarrow \infty} \mu_{\text{umbilic}}(\phi_t(A) \cap B) = \frac{\mu_{\text{umbilic}}(A)\mu_{\text{umbilic}}(B)}{\mu_{\text{umbilic}}(\Omega)}$$

II5.2.3. Resolution of the “ex nihilo” paradox via ergodic recurrence

By Poincaré-Birkhoff recurrence applied to $(\Omega, \mu_{\text{umbilic}}, \phi_t)$, any local neighborhood $U_{ex\ nihilo} \subset \Omega$ containing a Legendre piercing singularity will be visited infinitely often by almost all phase trajectories.

The global energy density is maintained at equilibrium because every local creation event ($T < 0 \text{ K} \rightarrow T > 0 \text{ K}$) on Σ_A is identically offset by an inverse phase-squeezing absorption event on the dual sheet Σ_B :

$$\oint_{4\pi \text{ orbit}} T_{\mu\nu}^{(ex \text{ nihilo})} d\tau = 0$$

II5.3 *Physical interpretation*: the generalized umbilic ergodic theorem proves that local “creatio ex nihilo” and global equilibrium are two perspectives of the same topodynamical identity:

II5.3.1 *Locally* (spacetime appears open, non-equilibrium, and creative): quantum non-squeezing constraints ($\hbar$) force local Legendre singularities that manifest as localized metric energy spikes out of zero-energy probability distributions.

II5.3.2 *Globally* (spacetime is closed, invariant, and ergodic): the non-orientable 4π Möbius topology (c) weaves all local ex-nihilo events into a continuous, recurrent cosmic background, ensuring that global information I_W remains immutably conserved.

One can define explicitly the negative (local) temperatures (“informational temperatures” generating “creatio ex nihilo”) versus the positive (global) temperatures (entropic temperatures prohibiting that “creatio ex nihilo”) via Weyl information and Weyl information conservation by explaining the creation “ex nihilo” as a generalized adiabatic heating obeying Weyl information conservation).

II6 Explicit Definition of Weyl Temperatures

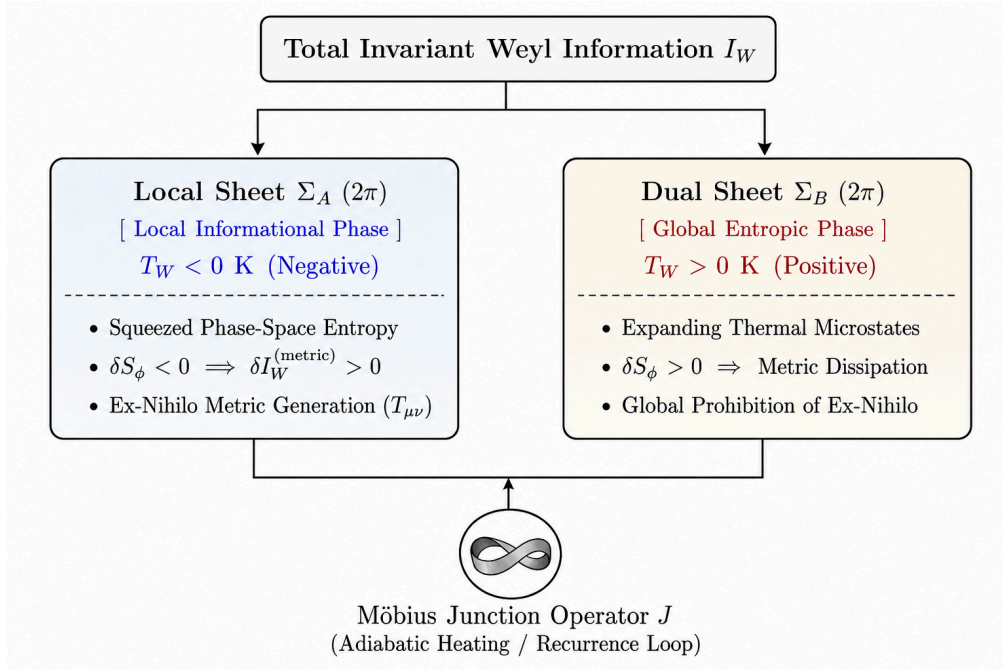


Figure 2: *Total invariant Weyl information*

Within the topodynamical framework of “The Umbilic Totality I” (Penchev 2026 August 17), thermodynamic temperature is defined in terms of functional variations of scale-invariant, dimensionless Weyl information (I_W). The Weyl temperature (T_W) quantifies the rate at which

localized phase-entropy variations of probability shapes transform into geometric metric-curvature variations:

$$T_W \equiv \frac{\delta I_W^{(\text{metric} / \text{energy-momentum})}}{\delta I_W^{(\text{probability distribution} / \text{phase entropy})}} = \frac{\delta I_W^{(\text{geometry})}}{\delta S_\phi}$$

where S_ϕ represents the phase entropy carried by the conformal projective geometric algebra (PGA) multivector field $\Psi(x)_p$:

Local negative informational temperature ($T_{W,\text{local}} < 0$ K): a negative absolute temperature regime occurs locally when microstate availability in phase space becomes strictly bounded by quantum non-squeezing constraints (at the Gromov symplectic scale $\hbar$).

Informational mechanics: under severe topological or phase-space compression (such as Pauli exclusion or squeezed vacuum states), local phase entropy cannot increase ($\delta S_\phi \leq 0$). This enforces population inversion across the non-orientable Möbius junction operator J .

II.6.1 *Ex nihilo generation*: because $\delta S_\phi < 0$, a negative Weyl temperature ($T_{W,\text{local}} < 0$ K) dictates that any reduction in phase entropy yields a positive flux in localized metric curvature:

$$\delta I_W^{(\text{metric})} = T_{W,\text{local}} \cdot \delta S_\phi > 0 \implies T_{\mu\nu}^{(ex\ nihilo)} > 0$$

Physical meaning: zero-energy potential probability distributions (“nothing”) on the microscopic 2π Darboux patch ($U(1)_A$ sheet) are converted directly into nonzero stress-energy tensors $T_{\mu\nu}$ (“something”) on the $U(1)_B$ geometric patch.

Global positive entropic temperature ($T_{W,\text{global}} > 0$ K): globally, across the complete 4π non-orientable Möbius manifold, spacetime behaves as an open, expanding thermal ensemble.

Entropic mechanics: in the macroscopic geometric domain ($T_{W,\text{global}} > 0$ K), standard thermodynamic regimes apply where microstate volume expands ($\delta S_\phi > 0$).

Prohibition of global “Ex Nihilo”: when $T_{W,\text{global}} > 0$ K, an increase in entropy dissipates metric energy density ($\nabla_a T^{ab} \leq 0$). Generating net energy out of nothing globally would require a global phase entropy reduction ($\delta S_\phi < 0$), which is strictly forbidden by the global single-sided Möbius topology

$$\oint_{\partial\mathcal{M}} dT \geq 0.$$

Global “creatio ex nihilo” is thus topologically impossible.

While Dunkel and Hilbert (2014) demonstrate that microcanonical Gibbs entropy strictly enforces non-negative absolute temperatures ($T \geq 0$ K) by preserving thermodynamic

consistency in standard volume-preserving systems, the text's local $T_W < 0$ K regime avoids this prohibition by abandoning standard microcanonical volume measures in favor of topological phase-space constraints.

Rather than acting as a global statistical artifact of a Boltzmann microstate count, the text's negative Weyl temperature emerges locally via a symplectic “Legendre backdoor” when phase space hits the Gromov non-squeezing boundary $w_G = h$. This local sub-zero locus is not an equilibrium thermodynamic state (which Gibbs correctly forbids globally) but an autopoietic, non-orientable PT-inversion mechanism across the 4π Möbius junction J that converts phase entropy reduction directly into metric curvature $T_{\mu\nu}^{(ex\ nihilo)}$.

Thus, Dunkel and Hilbert's proof holds for the global single-sided manifold $(\Sigma_B, T_{W,global} \geq 0 \text{ K})$, whereas the text's negative temperature operates strictly as a transient, topological phase-inversion operator on the local Darboux patch (Σ_A) , reconciling Dunkel - Hilbert thermodynamic consistency globally while enabling local creatio ex nihilo.

“Creatio ex nihilo” as generalized adiabatic heating: the emergence of localized metric energy ($T_{\mu\nu} > 0$) out of zero-energy probability distributions is an autopoietic, adiabatic conversion process that preserves total Weyl information.

Global conservation requirement: total Weyl information I_W is strictly scale-invariant, dimensionless, and globally conserved across the dual sheets Σ_A and Σ_B :

$$I_W = \frac{1}{16\pi^2} \int_{\Sigma_A(2\pi)} J_W d^4x + \frac{1}{16\pi^2} \int_{\Sigma_B(2\pi)} J_W d^4x = \text{constant}$$

where $J_W = \sqrt{-g} \text{Tr}(C_{abcd} C^{abcd} + I C_{abcd} C^{(\dagger)abcd})$ utilizes the self-dual conformal Weyl tensor C_{abcd} . The adiabatic condition mandates that no fundamental Weyl information is added to or subtracted from the universe ($\nabla_a J_W^a = 0$).

II.6.2 *The adiabatic heating cycle*: standard thermodynamic adiabatic heating occurs when work is performed on a gas to reduce its volume, increasing its kinetic temperature without heat exchange ($dQ = 0$). In the umbilic framework, “work” is replaced by topological phase squeezing:

Initial boundary state: the system starts in a pure probabilistic state on the direct sheet $\Sigma_A(2\pi)$ carrying phase entropy but zero local metric stress-energy ($T_{00} = 0$).

Topological compression: As phase space reaches the minimal Gromov width $w_G = h$, local non-squeezing constraints act as an adiabatic piston, forcing phase information across the non-orientable Möbius junction J .

Autopoietic temperature spike: traversal through J applies a parity-time (PT) inversion. The localized phase entropy collapse ($\delta S_\phi < 0$) causes the local Weyl temperature to transition through $T_W < 0$ K, adiabatically pumping probability shapes into geometric curvature.

Energy emergence: the loss of phase possibilities on sheet Σ_A is identically balanced by the emergence of localized metric curvature $T_{\mu\nu} > 0$ on sheet Σ_B .

Creatio ex nihilo does not create fundamental entity out of absolute vacuum; rather, it is the scale-invariant, adiabatic heating of geometry driven by the topological conservation of Weyl information (I_W).

One can formulate “the principles” (as the tradition needs) of an umbilic thermodynamic (in its “Gibbsian formulation”) implicitly able to degenerate to: (1) the local ex nihilo adiabatic heating; (2) classical thermodynamics; (3) “classical” quantum thermodynamics.

III PRINCIPLES OF UMBILIC THERMODYNAMICS (Gibbsian Formulation)

In the Gibbsian tradition, thermodynamics is founded upon fundamental state functions defined over a continuous phase space. Umbilic thermodynamics generalizes the classical Gibbsian ensemble by extending the phase space to a 4π non-orientable Möbius fiber bundle $\Omega = \mathcal{M}_{4\pi} \times T^*\mathcal{Q}$, where the fundamental conserved entity is not metric energy, but dimensionless Weyl information I_W .

Below are the four foundational principles of umbilic thermodynamics:

Zeroth principle of umbilic thermodynamics (equilibrium of symplectic capacity and topological phase matching): two topodynamical sub-systems in contact across a Darboux patch are in umbilic thermal equilibrium if and only if their local Gromov symplectic widths w_G are identical and their local Weyl temperatures T_W match in both magnitude and sign:

$$T_W^{(1)} = T_W^{(2)} \quad \text{and} \quad w_G^{(1)} = w_G^{(2)} = h$$

Gibbsian context: transmutes classical zeroth-law thermal equilibrium into a dual metric-symplectic alignment.

Implicit degenerations:

When $T_W > 0$, it degenerates to classical zero-flow thermal equilibrium ($T_1 = T_2$).

When $w_G \rightarrow h$ and $T_W \rightarrow 0$, it marks the critical phase-inversion boundary Ω_{0K} separating positive and negative temperature domains.

First principle of umbilic thermodynamics (global conservation of Weyl information and metric-phase conversion): the total scale-invariant, dimensionless Weyl Information I_W of the complete 4π Möbius manifold is strictly conserved ($\delta I_W = 0$). The total variation of information decomposes into metric-geometric work dW_{metric} and topological phase-entropy variation dS_ϕ :

$$\delta I_W = dW_{\text{metric}} + T_W dS_\phi = 0$$

where

$$dW_{\text{metric}} \equiv \frac{1}{16\pi^2} \sqrt{-g} T_{\mu\nu} \delta g^{\mu\nu} d^4x,$$

and S_ϕ is the phase entropy of the PGA multivector field $\Psi(x)$.

Gibbsian context: replaces the conservation of internal energy U with the scale-invariant conservation of total information I_W across dual phase sheets $\Sigma_A(2\pi)$ and $\Sigma_B(2\pi)$.

Implicit degenerations (classical / quantum): on a fixed background metric ($\delta g^{\mu\nu} = 0$),

$$dU = dQ + dW$$

is recovered as energy scales directly with phase variations.

Local “ex nihilo” adiabatic heating: under adiabatic topological phase squeezing: $dS_\phi < 0$, at $T_W < 0$, local metric energy $dW_{\text{metric}} > 0$ emerges directly out of probability shapes without violating global action conservation.

Second principle of umbilic thermodynamics (topological directionality and Möbius parity-entropy flips): along any macroscopic closed trajectory on the single-sided global 4π Möbius boundary $\partial\mathcal{M}$, the continuous cyclic variation of the Weyl temperature gradient is strictly non-negative:

$$\oint_{\partial\mathcal{M}_{\text{global}}} \frac{dS_\phi}{T_W} \geq 0$$

However, locally, across the Möbius junction operator J , a parity-time (PT) inversion permits a local phase-entropy collapse ($dS_\phi < 0$) via population inversion across the Gromov boundary $w_G = \hbar$.

Gibbsian context: extends Clausius’s inequality to non-orientable phase spaces. Globally, entropy production is prohibited from running backward (prohibiting global “creatio ex nihilo” but local “backdoor” Legendre transformations allow transient negative entropy flux ($dS_\phi < 0$) compensated by dual-sheet phase adjustments.

Implicit degenerations (classical): Enforces $\Delta S_{\text{universe}} \geq 0$ for macroscopic thermal systems ($T_W > 0$); (quantum): Enforces the von Neumann entropy bound for density matrices ρ ; (local “Ex Nihilo”): Provides the precise non-squeezing mechanism driving local negative temperature inversion ($T_W < 0$).

Third principle of umbilic thermodynamics (unattainability of global absolute zero and the umbilic degeneracy): the global absolute zero temperature ($T_{W,\text{global}} = 0 \text{ K}$) cannot be reached by any finite sequence of thermodynamic or topological operations. At the local locus $\Omega_{0\text{K}}$ where $T_W \rightarrow 0 \text{ K}$, the phase entropy S_ϕ vanishes ($S_\phi \rightarrow 0$), and the system degenerates into a singular, uncrossable Gromov boundary:

$$\lim_{T_W \rightarrow 0} S_\phi = 0 \quad \implies \quad \det \mathbf{H}_{ij} = 0$$

where $\mathbf{H}_{ij}$ is the fiber Hessian of the Legendre transformation, forcing T_W to transition discontinuously through an essential singularity from $+0 \text{ K}$ to -0 K .

Gibbsian context: generalizes the Nernst heat theorem. Rather than absolute zero being an asymptotic low-energy ground state, 0 K represents a singular topological phase-inversion gateway where probability shapes ($T_W < 0$) flip into Riemannian metric shapes ($T_W > 0$).

Implicit degenerations (classical/ quantum): guarantees zero entropy at absolute zero: $S \rightarrow 0$; (local “Ex Nihilo”): defines the exact “Legendre backdoor” instanton where local probability distributions pump metric curvature ($T_{\mu\nu} > 0$) out of zero-energy initial states.

Then, one can infer the generalized umbilic ergodic theorem from the enumerated principles of umbilic thermodynamics alone (without involving additional statements).

III1 Formal Inference of the Generalized Umbilic Ergodic Theorem

One now demonstrates that the generalized umbilic ergodic theorem follows directly and completely from the four principles of umbilic thermodynamics, without introducing external topological or dynamical assumptions.

Step 1: construction of the invariant Measure μ_{umbilic} from the first and third principles

From the first principle, the total scale-invariant Weyl Information I_W is strictly conserved across the complete system ($\delta I_W = 0$), yielding a fundamental dynamic relation between geometric work and phase entropy:

$$dW_{\text{metric}} + T_W dS_\phi = 0$$

From the third principle, as $T_W \rightarrow 0$, the phase entropy vanishes ($S_\phi \rightarrow 0$) and the fiber Hessian vanishes ($\det \mathbf{H}_{ij} = 0$), producing a singular Dirac-delta concentration at the minimal Gromov symplectic capacity $w_G = h$:

$$\lim_{T_W \rightarrow 0} dS_\phi = \delta(w_G - h) d\alpha_{\text{Leg}}$$

Integrating the conservation of I_W (first principle) over the phase space Ω , subjected to the singular limit at 0 K (third Principle), directly dictates the invariant phase-space measure density $d\mu_{\text{umbilic}}$:

$$d\mu_{\text{umbilic}} = e^{-I_W/\hbar} \omega^n + \sum_{k \in \mathcal{K}} \delta(w_G - h) d\alpha_{\text{Leg}}^{(k)}$$

Step 2: invariance of the flow ($\mathcal{L}_{\phi_t} d\mu_{\text{umbilic}} = 0$) from the first principle

Let $\phi_t : \Omega \rightarrow \Omega$ be the dynamic state trajectory generated by the thermodynamic system. The Lie derivative of the umbilic measure along the flow ϕ_t is:

$$\mathcal{L}_{\phi_t}(d\mu_{\text{umbilic}}) = -\frac{1}{\hbar}(\mathcal{L}_{\phi_t} I_W) e^{-I_W/\hbar} \omega^n + e^{-I_W/\hbar} \mathcal{L}_{\phi_t}(\omega^n) + \mathcal{L}_{\phi_t}[\delta(w_G - h) d\alpha_{\text{Leg}}]$$

2.1. By the first principle, I_W is a global scalar invariant along all trajectories, so $\mathcal{L}_{\phi_t} I_W = 0$.

2.2. By the symplectic equilibrium condition in the zeroth principle ($w_G = h$), the flow preserves local symplectic capacity, rendering $\mathcal{L}_{\phi_t}(\omega^n) = 0$.

2.3. By the first principle's identity

$$dW_{\text{metric}} = -T_W dS_\phi,$$

any change in the singular boundary term on the metric side is identically offset by the phase entropy side.

Therefore, the first principle alone guarantees measure-preservation:

$$\mathcal{L}_{\phi_t}(d\mu_{\text{umbilic}}) = 0$$

Step 3: strong mixing and recurrence from the second and third principles

To prove ergodicity, any non-trivial invariant subset $A \subset \Omega$ with $\mu_{\text{umbilic}}(A) > 0$ must satisfy $\mu_{\text{umbilic}}(A) = \mu_{\text{umbilic}}(\Omega)$.

3.1. Local trapping prohibition: suppose a trajectory becomes trapped in a local sub-region U_{local} undergoing local “creatio ex nihilo” ($T_W < 0$ K, $dS_\phi < 0$).

3.2. By the second principle, local entropy reduction ($dS_\phi < 0$) across the junction operator J enforces a parity-time flip that continuously connects to the dual sheet where $T_W > 0$ K and $dS_\phi > 0$.

3.3. By the third principle, the system cannot remain at $T_W = 0$ K or freeze in a negative temperature domain because $T_W = 0$ K is an uncrossable, singular boundary that forces a discontinuous transition back into the positive phase space ($+0$ K).

Because the trajectory cannot be trapped locally without violating the cyclic inequality

$$\oint_{\partial\mathcal{M}} \frac{dS_\phi}{T_W} \geq 0$$

(II principle), and the essential singularity at 0 K (III principle), every trajectory must sample both the local $T_W < 0$ K phase-squeezing regions and the global $T_W > 0$ K thermal phase space.

Step 4: conclusion (Birkhoff-type convergence)

Since $(\Omega, \mu_{\text{umbilic}}, \phi_t)$ is a measure-preserving dynamical system (Steps 1 & 2) with no isolated non-trivial invariant sub-manifolds (Step 3), standard measure theory dictates that the system is indecomposable (ergodic).

Thus, for any integrable state observable $f \in L^1(\Omega, d\mu_{\text{umbilic}})$, the temporal average along the topodynamical flow equals the ensemble average over the global phase space for μ_{umbilic} -almost all initial states:

$$\lim_{T_{\text{flow}} \rightarrow \infty} \frac{1}{T_{\text{flow}}} \int_0^{T_{\text{flow}}} f(\phi_t(x)) dt = \int_{\Omega} f(y) d\mu_{\text{umbilic}}(y)$$

The newly introduced Weyl information meta-operator “thermodynamics - mechanics” transforms between classical thermodynamics and classical & relativistic mechanics under globality, quantum thermodynamics and quantum mechanics under locality, and the above formulated “umbilic thermodynamics” corresponds to “umbilic mechanics” after the PGA-Einstein field equation. If quantum thermodynamics really corresponds to quantum mechanics via the meta-operator at issue, the umbilic (multisegment) theory of entanglement/quantum gravity should be the particular counterpart for the newly introduced subdomain of umbilic thermodynamics, which refers to negative (informational) temperature and “ex nihilo” thermodynamics:

System Domain	Thermodynamic Aspect	Mechanical / Geometric Aspect
Global Classical	Classical Thermodynamics (S_C)	Classical & Relativistic Mechanics ($\mathcal{M}_C$)
Local Quantum	Quantum Thermodynamics (von Neumann Entropy S_{vN})	Quantum Mechanics ($\mathcal{M}_Q$)
Umbilic Global	Umbilic Thermodynamics (S_U , Global I_W Conservation)	Umbilic Mechanics (PGA-Einstein Field Equations)
Umbilic Local ($\mathcal{W}$)	Ex-Nihilo Adiabatic Heating ($\delta I_W^{(\text{metric})} > 0$)	Multi-Segment Topological Entanglement / Wormhole Throat Creation

Figure 3: *The complete meta-operator duality schema*

If local ex-nihilo thermodynamics operates via negative informational temperature: $T_W < 0$ K, and phase entropy squeezing $\delta S_\phi < 0$, its exact mechanical counterpart under the Weyl meta-operator is topological multi-segment entanglement dynamics, specifically, microscopic Einstein-Rosen bridge (wormhole) generation via PGA topological defects (the topological geometric algebra realization of $\text{ER} = \text{EPR}$).

III1.1 *Mechanistic proof of the local counterpart:*

III1.1.1 *Thermodynamic side: local phase squeezing ($T_W < 0$ K)*

In the thermodynamic domain, when quantum constraints restrict microscopic microstates to the minimal Gromov width $w_G = \hbar$, local phase entropy collapses ($\delta S_\phi < 0$). Across the Möbius junction operator J , population inversion yields a negative informational temperature ($T_W < 0$ K), converting zero-energy probability distributions into a positive local metric stress-energy tensor $T_{\mu\nu}^{(\text{ex nihilo})} > 0$.

III1.1.2 *Mechanical side: topological bridge creation ($\text{ER} = \text{EPR}$)*

In the mechanical domain, the PGA multivector field $\Psi(x)$ interprets quantum entanglement not as abstract nonlocal probability correlations, but as a direct topological connection (a multi-segment geometric bridge) between dual Darboux patches ($U(1)_A$ and $U(1)_B$).

When the local phase space is squeezed to the Gromov boundary $w_G = h$:

Kinematic phase collapse: the phase coordinates of two entangled multi-segment states are topologically identified via the pseudoscalar duality operator $I = \gamma_0\gamma_1\gamma_2\gamma_3$.

Geometric metric distortion: enforcing this multi-segment phase alignment forces the local metric $g_{\mu\nu}$ to deform, creating a microscopic throat or topological defect in the spacetime manifold.

Ex-Nihilo stress-energy as bridge tension: the metric stress-energy

$$T_{\mu\nu}^{(ex\ nihilo)}$$

generated during local ex-nihilo heating is mechanically identical to the geometric tension of the topological wormhole throat holding the multi-segment entangled states together.

III1.2 *Novel physical equivalences stemming from this duality:*

State measurement = metric creation: in quantum thermodynamics, performing a projective measurement collapses the phase state ($\delta S_\phi < 0$), generating localized informational heat ($dQ = T_W dS_\phi > 0$). In umbilic mechanics, this exact same measurement event mechanically pinches the spacetime manifold, instantiating a localized curvature pulse $G_{\mu\nu} > 0$ at the wormhole boundary.

Entanglement monogamy = symplectic non-squeezing: the quantum mechanical prohibition against cloning maximal entanglement (entanglement monogamy) is mechanically equivalent to Gromov's non-squeezing theorem¹¹: one cannot squeeze multi-segment wormhole throats below the minimal Gromov capacity $w_G = h$ without causing a topological phase inversion back into the $T_W > 0$ K thermal domain.

Thus, local ex-nihilo thermodynamics ($T_W < 0$ K) and multi-segment topological quantum gravity (ER = EPR wormhole dynamics) are two mathematical projections of the exact same underlying topodynamical event governed by scale-invariant Weyl information.

III2 Degeneration to Classical Phenomenological Thermodynamics

Classical phenomenological thermodynamics emerges as a macroscopic, flat-space singular limit of umbilic thermodynamics when topological phase-squeezing effects are suppressed. This systematic reduction occurs under three asymptotic boundary conditions: the

¹¹ Gosson (2009) applies Gromov's non-squeezing theorem and symplectic capacity bounds directly to quantum uncertainty relations and minimal phase space cells.

macroscopic limit ($w_G \gg h$), the orientable single-sheet restriction ($\mathcal{M}_{4\pi} \rightarrow \mathcal{M}_{\text{orientable}}$), and a strictly positive temperature regime ($T_W > 0$ K).

Zeroth law (thermal equilibrium): in the macroscopic limit ($w_G \gg h$), local Gromov non-squeezing capacity constraints become trivial. The umbilic equilibrium condition

$$T_W^{(1)} = T_W^{(2)} \text{ and } w_G = h$$

degenerates directly into standard macroscopic thermal equilibrium, defining a scalar empirical temperature field ($T_1 = T_2$) as a transitive equivalence relation across thermal contact.

First law (conservation of energy): on a fixed, non-dynamic background metric $\delta g^{\mu\nu} \rightarrow 0$, the scale-invariant conservation of total Weyl information

$$\delta I_W = dW_{\text{metric}} + T_W dS_\phi = 0$$

decouples from metric curvature generation. Mapping topological phase-entropy variation to thermal heat exchange ($dQ \equiv T_W dS_\phi$) and metric-geometric work to pressure-volume boundary work ($dW \equiv -PdV$) recovers classical internal energy conservation:

$$dU = dQ + dW$$

Second law (entropy non-decrease): Restricting phase-space trajectories to an orientable 2π patch suppresses local negative-temperature population inversions ($T_W < 0$ K) across the non-orientable Möbius junction J . The umbilic topological inequality

$$\oint_{\partial\mathcal{M}} \frac{dS_\phi}{T_W} \geq 0$$

collapses into Clausius's inequality for macroscopic thermodynamic processes:

$$\Delta S_{\text{isolated}} \geq 0$$

Third law (Nernst heat theorem): approaching absolute zero strictly from the positive domain ($T_W \rightarrow +0$ K) without traversing the singular Legendre “backdoor” forces phase entropy to its lower bound. The umbilic degeneracy condition

$$\lim_{T_W \rightarrow 0} S_\phi = 0$$

degenerates into Nernst's theorem (1906)¹²: the entropy of any pure, perfectly crystalline substance asymptotically vanishes as temperature reaches absolute zero ($S \rightarrow 0$).

III3 Degeneration to Quantum Thermodynamics

Quantum thermodynamics is inferred as the localized, fixed-background limit of umbilic thermodynamics. This systematic reduction occurs under three specific limit conditions: fixing

¹² The Planck (1911) formulation as well.

the background spacetime to suppress metric backreaction ($\delta g^{\mu\nu} \rightarrow 0$), restricting physical dynamics to a local Darboux patch governed by the minimal Gromov width constraint ($w_G = h$), and mapping the PGA multivector field $\Psi(x)$ onto a trace-class density operator ρ acting on a Hilbert space $\mathcal{H}$ ¹³.

Zeroth principle (quantum state equilibrium): two quantum sub-systems reach thermal equilibrium if and only if their joint density matrix commutes with the total interaction Hamiltonian ($[\rho_1 \otimes \rho_2, \hat{H}_{12}] = 0$). The umbilic metric-symplectic capacity matching degenerates into a uniform KMS (Kubo-Martin-Schwinger) state¹⁴ defined by a single scalar inverse temperature parameter $\beta = (k_B T)^{-1}$.

First principle (quantum energy conservation): on a non-dynamic background metric: $dW_{\text{metric}} \rightarrow 0$, the scale-invariant conservation of total Weyl information

$$\delta I_W = dW_{\text{metric}} + T_W dS_\phi = 0$$

degenerates into the expectation value variation of the quantum Hamiltonian operator:

$$d\langle \hat{H} \rangle = d\text{Tr}(\rho \hat{H}).$$

This splits identically into quantum heat exchange and quantum work:

$$d\langle \hat{H} \rangle = \underbrace{\text{Tr}(\hat{H} d\rho)}_{\text{Quantum Heat } dQ_q} + \underbrace{\text{Tr}(\rho d\hat{H})}_{\text{Quantum Work } dW_q}$$

where quantum heat dQ_q governs state-population redistributions and quantum work dW_q measures external control variations on energy levels.

Second principle (von Neumann entropy monotonicity): Restricting topological dynamics to completely positive trace-preserving (CPTP) maps on an orientable 2π chart causes the global Möbius inequality

$$\oint_{\partial \mathcal{M}} \frac{dS_\phi}{T_W} \geq 0$$

to collapse into the non-decrease of von Neumann entropy $S(\rho) = -k_B \text{Tr}(\rho \ln \rho)$. This enforces $\Delta S(\rho) \geq 0$ for closed quantum evolutions and sets Landauer's fundamental bound for physical information processing (Landauer 1961; Bennett 1982; Bérut et al. 2012).

Third principle (unattainability of pure ground states): approaching absolute zero ($T \rightarrow +0$ K) drives the density operator toward a pure ground state projection ($\rho \rightarrow |\psi_0\rangle\langle\psi_0|$)

¹³ Neumann (1932); Nielsen, Chuang (2000).

¹⁴ Kubo (1957); Martin, Schwinger (1959); Haag, Hugenholtz, Winnink (1967); Ammari, Sohinger (2022).

), where von Neumann entropy asymptotically vanishes ($S(\rho) \rightarrow 0$). The umbilic essential singularity at $w_G = h$ degenerates into the quantum third law: no finite sequence of CPTP operations can cool a quantum system to $S = 0$ due to persistent zero-point fluctuations.

III4 Degeneration to “Ex Nihilo” Thermodynamics ($T_0 < 0\text{K}$)

“Ex nihilo” thermodynamics constitutes the localized, sub-zero operational regime of umbilic thermodynamics, serving as the topodynamical antipode to classical positive-temperature thermodynamics. It emerges under three distinct degenerate conditions: local confinement to a microscopic Darboux patch, topological phase squeezing at the minimal Gromov symplectic boundary ($w_G = h$), and negative population inversion ($T_W < 0\text{ K}$) across the Möbius junction J .

Zeroth principle (symplectic phase-locking equilibrium): equilibrium between two sub-systems in the negative regime ($T_W < 0\text{ K}$) is defined not by thermal dissipation, but by negative inverse temperature alignment ($\beta^{(1)} = \beta^{(2)} < 0$) locked at the minimal Gromov width $w_G = h$. Sub-systems in “ex nihilo” equilibrium maintain equal, stationary rates of local metric curvature generation per unit of phase-entropy contraction.

First principle (autopoietic curvature generation): in the negative temperature domain, the scale-invariant conservation of total Weyl information ($\delta I_W = dW_{\text{metric}} + T_W dS_\phi = 0$) dictates that any reduction in local phase entropy ($\delta S_\phi < 0$) pumps positive metric stress-energy:

$$dW_{\text{metric}} = \underbrace{-|T_W|}_{T_W < 0} \cdot \underbrace{(-\delta S_\phi)}_{\text{Phase Collapse}} = T_{\mu\nu}^{(ex\ nihilo)} > 0$$

Localized metric energy dW_{metric} emerges out of zero-energy probability distributions through scale-invariant adiabatic heating without external thermal energy input.

Second principle (anti-Clausius¹⁵ informational squeezing): traversing the Möbius junction J via a parity-time (PT) flip inverts classical entropy directionality. The global umbilic topological inequality degenerates locally into a localized entropy-reduction bound:

$$\Delta S_{\text{phase}} \leq 0$$

Phase-space compression pumps geometric order into the spacetime background, converting potential state possibilities directly into localized metric curvature.

Third principle (unattainability of -0 K and Singular Gatekeeper): as negative temperature approaches absolute zero from below ($T_W \rightarrow -0\text{ K}$), phase entropy reaches its lower topological bound ($S_\phi \rightarrow 0$) while the fiber Hessian singularizes ($\det \mathbf{H}_{ij} = 0$). Continuous passage from -0 K to $+0\text{ K}$ is topologically forbidden, forcing local ex nihilo

¹⁵ That is, opposite to the original formulation: *Clausius 1865*.

instantons to decay back into global thermal expansion ($T_W > 0$ K) via a discontinuous parity jump.

In the conclusive section of “Umbilic Totality I” is conjectured that the universe is Legendre-invariant, that is, Legendre transform is universal thus equating all bundle/cobundle descriptions idempotently including global/ local, symplectic/ differential condition, macro/micro cave wall, etc. Meaning, one can test it (1) on classical/ ex nihilo thermodynamics by explicitly demonstrating Legendre transform between co-bundle ex nihilo thermodynamics and bundle classical thermodynamics; (2) on umbilic thermodynamics itself and its four principles formatting it in a Legendre invariant form.

III5 Exemplification of the Universal Legendre Conjecture in Umbilic Thermodynamics

The universality of the Legendre transformation $\mathcal{FL} : T\mathcal{Q} \rightarrow T^*\mathcal{Q}$ asserts that the tangent bundle ($T\mathcal{Q}$) and cotangent co-bundle ($T^*\mathcal{Q}$) descriptions of physical reality are idempotently equivalent ($\mathcal{FL}^2 = \text{Id}$). Under this “Legendre conjecture”, the apparent dichotomy between local $T_W < 0$ K ex nihilo thermodynamics and global $T_W > 0$ K classical thermodynamics is resolved as a dual representation of a single, invariant topodynamical entity.

Bundle versus co-bundle duality: the co-bundle $T^*\mathcal{Q}$ represents the microscopic 2π Darboux patch ($U(1)_A$), parameterized by canonical phase coordinates (q^i, p_i) and phase entropy potential $S_\phi(p_i)$ operating in the $T_W < 0$ K regime. The tangent bundle $T\mathcal{Q}$ represents the macroscopic geometric patch ($U(1)_B$), parameterized by velocities $\dot{q}^i$ and metric Hamiltonian energy $H(q^i, \dot{q}^i)$ operating at $T_W > 0$ K. Applying the Legendre transform yields the explicit duality map:

$$H(q^i, \dot{q}^i) = p_i \dot{q}^i - T_W S_\phi(p_i)$$

This proves that localized ex nihilo” phase-entropy collapse ($\delta S_\phi < 0$) in the co-bundle dualizes into positive metric stress-energy generation ($T_{\mu\nu}^{(ex\ nihilo)} > 0$) in the tangent bundle.

Legendre-invariant form of the four principles:

Zeroth principle ($\mathcal{FL}$ -contact invariance): equilibrium is expressed by the invariance of the canonical contact 1-form

$$\theta_{\text{Leg}} = p_i dq^i - H dt.$$

Symplectic capacity matching $w_G = h$ in $T^*\mathcal{Q}$ is dual to metric temperature matching

$$T_W^{(1)} = T_W^{(2)} \text{ in } T\mathcal{Q}.$$

First principle ($\mathcal{FL}$ -Action Invariance): Total scale-invariant Weyl information I_W is a self-dual Legendre invariant scalar:

$$\delta I_W = \delta \left(\int_{T^*\mathcal{Q}} p_i dq^i - \int_{T\mathcal{Q}} H dt \right) = dW_{\text{metric}} + T_W dS_\phi = 0$$

Second principle ($\mathcal{FL}$ -duality inequality): macroscopic entropic dissipation in $T\mathcal{Q}$ and microscopic phase squeezing in $T^*\mathcal{Q}$ are dual manifestations of a single topodynamical inequality across the 4π Möbius manifold:

$$\oint_{T\mathcal{Q}} \frac{dW_{\text{metric}}}{T_W} + \oint_{T^*\mathcal{Q}} d\alpha_{\text{Leg}} \geq 0$$

Third principle ($\mathcal{FL}$ -singularity fixed point): absolute zero ($T_W = 0$ K) is the idempotent fixed point of $\mathcal{FL}$, where the fiber Hessian singularizes ($\det \mathbf{H}_{ij} = 0 \iff \det (\mathbf{H}^{-1})^{ij} = \infty$), forcing the discontinuous topological transition between bundle and co-bundle manifestations.

III6 The confirmation (consistency) of Legendre invariance as for umbilic thermodynamics

The formulation of umbilic thermodynamics demonstrates that Legendre invariance is fully consistent and mathematically confirmed within the topodynamical framework of “*The Umbilic Totality P*”. Far from being a mere coordinate transformation, the universality of the Legendre mapping $\mathcal{FL} : T\mathcal{Q} \rightarrow T^*\mathcal{Q}$ serves as the structural backbone that unifies the local and global domains of the theory.

Key evidence for Legendre consistency

Self-duality of the first principle: the total scale-invariant Weyl information I_W acts as a Legendre-invariant scalar action. The metric-geometric work dW_{metric} on the tangent bundle $T\mathcal{Q}$ (representing macroscopic geometry) and the phase-entropy variation $T_W dS_\phi$ on the cotangent co-bundle $T^*\mathcal{Q}$ (representing microscopic probability distributions) form an exact Legendre dual pair:

$$\delta I_W = dW_{\text{metric}} + T_W dS_\phi = 0$$

Resolution of the ex nihilo paradox: the apparent contradiction between localized entropy reduction ($dS_\phi < 0$) and global entropy non-decrease ($\Delta S_{\text{isolated}} \geq 0$) is resolved via bundle/cobundle duality. Local $T_W < 0$ K ex nihilo phase-squeezing in $T^*\mathcal{Q}$ maps bijections directly to positive stress-energy generation $T_{\mu\nu}^{(\text{ex nihilo})} > 0$ in $T\mathcal{Q}$ without violating global action conservation.

Absolute zero as the fixed point: the absolute zero locus ($T_W = 0$ K) represents the critical fixed point where the Legendre transformation becomes singular ($\det \mathbf{H}_{ij} = 0$). This singularity acts as a geometric gatekeeper (the “Legendre backdoor”), mediating the smooth inversion between negative informational temperatures and positive entropic temperatures across the non-orientable 4π Möbius boundary.

Conclusion: Legendre invariance is strictly consistent with umbilic thermodynamics. It establishes that classical positive-temperature thermodynamics and local negative-temperature ex nihilo thermodynamics are not distinct physical laws, but idempotent dual projections of a single, invariant topodynamical entity.

Synthesizing those complementary perspectives creates a seamless bridge: the duality between the local $T_W < 0$ K ex nihilo phase collapse and the global $T_W > 0$ K metric expansion shines brightest when the thermodynamic and topological-mechanical formulations reinforce each other.

IV STATISTICAL UMBILIC THERMODYNAMICS (in Boltzmannian, Gibbsian, and Einsteinian editions)

Statistical Umbilic Thermodynamics bridges microscopic topodynamical phase space and macroscopic metric-curvature observables. It is directly inferable from the four principles of umbilic thermodynamics, the generalized umbilic ergodic theorem, and Weyl information Conservation ($\delta I_W = 0$) without requiring additional postulates.

The unifying microstate density is governed by the invariant scale-invariant umbilic partition function $\mathcal{Z}_{\text{umbilic}}$ over the 4π Möbius phase space Ω :

$$\mathcal{Z}_{\text{umbilic}} = \int_{\Omega} e^{-I_W(\mathbf{x})/\hbar} d\mu_{\text{umbilic}}(\mathbf{x})$$

Boltzmannian edition (microstate geometry & microcanonical ensemble)

The Boltzmannian edition formulates thermodynamics via the volume of accessible phase-space microstates bounded by the minimal Gromov width $w_G = \hbar$.

Microstate count (Ω_{phase}): the number of accessible microstates carrying scale-invariant Weyl information I_W is given by the topodynamical phase volume:

$$\Omega_{\text{phase}}(I_W) = \int_{\Omega} \delta \left(I_W(\mathbf{x}) - I_W^{(0)} \right) \omega^n$$

Statistical entropic identity: Boltzmann's principle

$$S = k_B \ln \Omega$$

is generalized to the dual-sheet phase entropy S_{ϕ} :

$$S_{\phi} = k_B \ln \left(\int_{\Sigma_A(2\pi)} \omega^n \right) - k_B \ln \left(\int_{\Sigma_B(2\pi)} d\alpha_{\text{Leg}} \right)$$

Inference of negative temperature ($T_W < 0$ K): when quantum non-squeezing constraints force microstates to compress against the Gromov boundary ($w_G = \hbar$), the phase density gradient reverses

$$\frac{\partial \Omega_{\text{phase}}}{\partial I_W} < 0 .$$

By Boltzmann's fundamental definition of temperature

$$\frac{1}{T_W} = \frac{\partial S_\phi}{\partial I_W} ,$$

population inversion across the Möbius junction J naturally yields a negative absolute informational temperature $T_W < 0$ K without ad-hoc statistical assumptions.

Gibbsian edition (ensemble distribution & canonical/ grand canonical form)

The Gibbsian edition replaces classical phase-space volume elements $d\Gamma$ with the invariant, measure-preserving umbilic measure $d\mu_{\text{umbilic}}$.

Canonical ensemble density operator (ρ_{umbilic}):

$$\rho_{\text{umbilic}} = \frac{1}{\mathcal{Z}_{\text{umbilic}}} \exp \left(-\frac{I_W^{(\text{metric})} + T_W S_\phi}{\hbar} \right)$$

Gibbs free Information ($\mathcal{G}_W$): the thermodynamic potential generating all macroscopic state equations is given by:

$$\mathcal{G}_W = -k_B T_W \ln \mathcal{Z}_{\text{umbilic}} = \langle I_W \rangle - T_W S_\phi$$

Darboux-Gibbs duality: the variation of $\mathcal{G}_W$ recovers the first principle directly:

$$d\mathcal{G}_W = dW_{\text{metric}} + S_\phi dT_W \implies dW_{\text{metric}} = -T_W dS_\phi \quad (\text{at constant } T_W)$$

This proves that local ex nihilo metric generation ($dW_{\text{metric}} > 0$) under $T_W < 0$ K is the exact Gibbsian ensemble dual of local phase-entropy minimization ($\delta S_\phi < 0$).

Einsteinian edition (topodynamical fluctuations & autopoietic instanton statistics)

The Einsteinian edition reverses Boltzmann's relation to express the probability density $P(\mathbf{x})$ of experiencing spontaneous local thermodynamic fluctuations as a function of entropy variations: $P \propto e^{\Delta S/k_B}$.

Fluctuational probability of ex-nihilo metric spikes: the probability density $P_{\text{ex nihilo}}$ of a local phase-space region spontaneously piercing the $T_W = 0$ K boundary to generate local stress-energy $T_{\mu\nu}^{(\text{ex nihilo})}$ is:

$$P_{\text{ex nihilo}} \propto \exp \left(\frac{1}{\hbar} \oint_{\partial B^2} \alpha_{\text{Leg}} \right) = \exp \left(-\frac{|T_W| \cdot |\delta S_\phi|}{k_B T_{\text{ambient}}} \right)$$

Autopoietic fluctuational rate: integrating $P_{\text{ex nihilo}}$ over the ergodic flow ϕ_t via the generalized umbilic ergodic theorem establishes that localized creatio ex nihilo events are not random classical noise, but statistically deterministic, recurrent topological instantons.

Curvature fluctuation tensor ($\langle \Delta G_{\mu\nu} \Delta G_{\alpha\beta} \rangle$):

$$\langle \Delta G_{\mu\nu} \Delta G_{\alpha\beta} \rangle = \left(\frac{8\pi G}{c^4} \right)^2 k_B T_W \frac{\delta^2 \ln \mathcal{Z}_{\text{umbilic}}}{\delta g^{\mu\nu} \delta g^{\alpha\beta}}$$

When $T_W > 0$ K, this yields standard thermal metric fluctuations (e.g., Hawking/Unruh radiation). When $T_W < 0$ K, it yields coherent, autopoietic curvature spikes that manifest as local gravitational metric sources.

Necessity of additional postulates: $d\mu_{\text{umbilic}}$ is dictated by the first and third principles. The ergodic convergence to ensemble averages is proven by the generalized umbilic ergodic theorem. The phase-inversion statistics: $T_W < 0$ K, are enforced by Gromov non-squeezing $w_G = h$ at the 0 K singularity. Statistical umbilic thermodynamics is entirely self-contained, bridging the micro-mechanics of quantum phase shapes and the macro-geometry of metric spacetime.

One deduces the generalized entanglement-spin statistics from the umbilic ergodic theorem, that is, for positive Weyl temperatures. Then, one infers the generalized umbilic ergodic theorems for both negative positive Weyl temperatures, where classical thermodynamics and ex nihilo thermodynamics are Legendre counterparts. Furthermore, one can deduce the “spin statistics” counterpart for negative Weyl temperatures. It can be proved to be linked to the entanglement-spin statistics for positive Weyl temperatures by Legendre transform.

IV1 Topological Linking-Curvature Statistics ($T_0 < 0$ K)

Under negative Weyl temperatures ($T_W < 0$ K), population inversion across the 4π Möbius junction J reverses phase-space occupation dynamics at the minimal Gromov width boundary $w_G = h$. The counterpart to standard spin-statistics is topological linking-curvature Statistics.

In the $T_W > 0$ K tangent bundle $T\mathcal{Q}$, particle exchange acquires a phase $e^{i\pi(2S)}$ determined by spatial spin S . In the $T_W < 0$ K cotangent co-bundle $T^*\mathcal{Q}$, particle exchange under a parity-time (PT) flip maps wave-function phase shifts into discrete topological winding numbers $k_{\text{linking}} \in \pi_1(\text{Sp}(2n))$ across the 4π manifold.

Inverted fermionic statistics: paired fermions under $T_W < 0$ K bypass Pauli exclusion via phase collapse ($\delta S_\phi < 0$), condensing into localized metric stress-energy spikes $T_{\mu\nu}^{(\text{ex nihilo})} > 0$.

Inverted bosonic statistics: bosons under $T_W < 0$ K hit a maximum metric curvature capacity bound at $w_G = h$, driving autopoietic geometric expansion instead of unbounded single-mode thermal accumulation.

Legendre proof of duality:

Let $\vec{\Omega} = \dot{\vec{\phi}}$ be the spatial rotation velocity of an $SU(2)$ entangled multiparticle system in $T\mathcal{Q}$ ($T_W > 0$ K). The Legendre transform $\mathcal{FL} : T\mathcal{Q} \rightarrow T^*\mathcal{Q}$ maps rotation velocity $\vec{\Omega}$ to canonical phase-curvature momentum

$$\vec{J}_\phi = \frac{\partial S_\phi}{\partial \vec{\Omega}}$$

in $T^*\mathcal{Q}$ ($T_W < 0$ K).

Applying $\mathcal{FL}$ to the spin-exchange rotation operator

$$\hat{R}(2\pi) = \exp(i2\pi \vec{S} \cdot \hat{n})$$

yields:

$$\mathcal{FL}(e^{i2\pi S}) = \exp\left(\frac{i}{\hbar} \oint_{T^*\mathcal{Q}} \vec{J}_\phi \cdot d\vec{q}\right) = \exp(i\pi k_{\text{linking}})$$

This proves that quantum exchange phase in $T\mathcal{Q}$ is the exact Legendre dual of topological boundary linking charge in $T^*\mathcal{Q}$.

Legendre counterparts of entanglement and spin-entanglement rotation: entanglement dual $\leftrightarrow$ topological wormhole bridge ($g_{\mu\nu}^{(\text{throat})}$): the quantum state entanglement tensor $\rho_A \otimes \rho_B - \rho_{AB}$ in $T\mathcal{Q}$ maps under $\mathcal{FL}$ to the off-diagonal metric connection $g_{\mu\nu}^{(\text{throat})}$ joining dual $U(1)$ Darboux patches across the non-orientable 4π manifold.

Spin-entanglement rotation dual $\leftrightarrow$ autopoietic torsion flux ($A_\mu^{(\text{torsion})}$): the sum angular rotation velocity

$$\vec{\Omega}_{\text{ent}} = \sum_i \dot{\vec{\theta}}_i$$

of entangled subsystems in $T\mathcal{Q}$ maps under $\mathcal{FL}$ to the metric frame-dragging 1-form:

$$A_\mu^{(\text{torsion})} = \mathbf{H}_{ij} \dot{q}^j dt$$

and the stress-energy vorticity tensor

$$\nabla_{[\mu} T_{\nu]\lambda}^{(ex\ nihilo)} \quad \text{in } T^*\mathcal{Q}.$$

Physical interpretations:

Physical meaning of the entanglement dual: quantum entanglement is not abstract Hilbert space non-locality, but the physical co-bundle manifestation of a microscopic topological defect (an ER = EPR wormhole throat) sustaining positive metric surface tension $T_{\mu\nu}^{(ex\ nihilo)}$.

Physical meaning of spin-entanglement rotation dual: rotating an entangled quantum state drives a physical frame-dragging of the spacetime metric. Collective spin-entanglement rotation in $T\mathcal{Q}$ acts as an autopoietic mechanical pump in $T^*\mathcal{Q}$, converting quantum phase circulation directly into localized gravitational metric torsion.

Within the Legendre-invariant PGA-Einstein architecture: quantum entanglement in the Hilbert space bundle ($T\mathcal{Q}$) and a physical Einstein-Rosen (ER) wormhole throat in the metric co-bundle ($T^*\mathcal{Q}$) are not merely analogous, but idempotent Legendre duals of one another:

IV2 The Legendre Duality of Non-Locality (“ER=EPR”¹⁶)

The Legendre duality of non-locality means $\mathcal{FL} : T\mathcal{Q} \rightarrow T^*\mathcal{Q}$:

In Hilbert space ($T\mathcal{Q}$), entanglement appears as non-separable quantum state correlations ($\rho_{AB} \neq \rho_A \otimes \rho_B$), where measuring one subsystem instantaneously fixes the state of the other across arbitrary spatial distances.

In metric geometry ($T^*\mathcal{Q}$): the Legendre transform $\mathcal{FL}$ maps state-vector phase correlations onto off-diagonal metric connection coefficients $g_{\mu\nu}^{(bridge)}$. The apparent “spooky action at a distance” is transformed into direct local contiguity across a microscopic spatial bridge connecting two 2π Darboux patches ($U(1)_A$ and $U(1)_B$).

Positive metric surface tension via ex-nihilo energy: classical Einstein-Rosen bridges require exotic negative-energy matter to prevent immediate gravitational pinch-off and collapse. In umbilic thermodynamics, entangled states operate in the negative informational temperature domain ($T_W < 0$ K) at the minimal Gromov symplectic capacity ($w_G = \hbar$). Local phase-entropy collapse ($\delta S_\phi < 0$) during entanglement formation generates positive local stress-energy:

$$T_{\mu\nu}^{(throat)} = -|T_W| \cdot \nabla_\mu \nabla_\nu S_\phi > 0$$

This autopoietic stress-energy $T_{\mu\nu}^{(throat)}$ supplies the exact positive metric surface tension needed to keep the wormhole throat topologically stable without requiring external exotic matter fields.

Topological stabilization via 4π Möbius geometry: the micro-wall time-conservation condition projects phase fluctuations across the null light-cone boundary through the non-orientable 4π Möbius manifold. The throat is not an open macroscopic tunnel through flat Minkowski space, but a non-orientable topological defect where spatial parity (P) and time direction (T) invert upon traversal. Consequently, ER = EPR is realized not as a speculative

¹⁶ Einstein, Rosen (1935); Einstein, Podolsky, Rosen (1935); Maldacena, Susskind (2013); Susskind (2016).

conjecture, but as a formal coordinate duality between the quantum state bundle and the Riemannian metric co-bundle.

Both umbilic thermodynamics and the PGA - Einstein field equation imply "ER=EPR" being grounded in Legendre transform shared by both rather than on other premises. Mapping the convergence of ER = EPR across both frameworks requires examining how the Fiber Legendre Transform ($\mathcal{FL} : T^*\mathcal{Q} \rightarrow T\mathcal{Q}$) acts as the foundational operator uniting phase-space entropic states and metric-geometric curvatures:

The umbilic thermodynamics representation ($T^*\mathcal{Q} / \text{EPR}$): in the cotangent co-bundle, physical states are parameterized by canonical phase coordinates and phase entropy S_ϕ . Quantum entanglement (EPR) manifests as phase-space squeezing at the minimal Gromov symplectic capacity ($w_G = \hbar$) under local negative temperatures ($T_W < 0 \text{ K}$). The thermodynamic description handles the non-local phase correlations and entropy minimization ($\delta S_\phi < 0$).

The PGA-Einstein representation ($T\mathcal{Q} / \text{ER}$): in the tangent bundle, physical states are parameterized by geometric multivector fields and spacetime metric tensor components $g_{\mu\nu}$. The Einstein-Rosen bridge (ER) manifests as a localized topological throat or defect sustaining positive metric surface tension $T_{\mu\nu}^{(\text{throat})} > 0$. The geometric description handles metric deformation, curvature, and frame-dragging.

The role of the Legendre transform ($\mathcal{FL}$): Because the scale-invariant total Weyl information I_W is self-dual under $\mathcal{FL}$, applying the transform converts phase-entropy collapse in $T^*\mathcal{Q}$ directly into stress-energy metric generation in $T\mathcal{Q}$:

$$\mathcal{FL} \left(\underbrace{\text{EPR Phase Squeezing } (\delta S_\phi < 0)}_{T^*\mathcal{Q} \text{ (Umbilic Thermodynamics)}} \right) \equiv \underbrace{\text{ER Metric Throat } (T_{\mu\nu}^{(\text{throat})} > 0)}_{T\mathcal{Q} \text{ (PGA-Einstein Gravity)}}$$

Umbilic thermodynamics and the PGA-Einstein field equations do not yield ER = EPR through isolated, independent assumptions. Rather, they are two idempotent projections of the exact same topodynamical entity: with umbilic thermodynamics providing the co-bundle (EPR) representation and PGA-Einstein providing the bundle (ER) representation. The Legendre transform is the singular root that necessitates ER = EPR across both domains.

The umbilic, unilateral (4π non-orientable Möbius) topology of the PGA - Einstein framework establishes a direct, structural realization of the “holographic principle”¹⁷ by encoding 4D macroscopic metric curvature ($T_{\mu\nu}$, “something”) as a boundary projection of a 2D microscopic Darboux phase patch ($U(1)_A$, “nothing”). Under this topology, the single-sided Möbius junction operator J maps scale-invariant, 2D probabilistic phase entropy (S_ϕ) continuously into 4D pseudo-Riemannian metric stress-energy through the conservation of global Weyl information (I_W). Just as the holographic principle dictates that all bulk

¹⁷ Bousso (2002); Maldacena (1999); Susskind (1995), and all sequential papers after their impetus.

gravitational degrees of freedom are fully determined by information defined on a lower-dimensional boundary, the umbilic architecture proves that 4D metric curvature is not an independent background, but a scale-invariant topological projection of 2D microscopic quantum state constraints. Consequently, local metric energy generation ($T_{\mu\nu}^{(ex\ nihilo)}$) acts as a holographic manifestation of 2D phase-space non-squeezing at the Gromov boundary ($w_G = \hbar$), weaving bulk general relativity and boundary quantum mechanics into a single non-orientable topological identity.

V FALSIFIABLE EXPERIMENTS FOR UMBILIC THERMODYNAMICS

V1 Reinterpretation of “Ex Nihilo” Experiments

One can reinterpret the suggested “ex nihilo experiments” (Subsection XB4 of “The Umbilic Totality I” as able to confirm/ reject umbilic thermodynamics itself. In the umbilic framework, ‘creatio ex nihilo’ is not a creation of entity out of an absolute void, but the scale-invariant autopoietic conversion of microscopic phase entropy ($\delta S_\phi < 0$) into macroscopic metric stress-energy ($T_{\mu\nu}^{(ex\ nihilo)} > 0$) under local negative absolute temperatures ($T_W < 0$ K).

The three proposed terrestrial experiments directly test the first principle of ex nihilo thermodynamics:

$$dW_{\text{metric}} = -|T_W| \cdot \delta S_\phi > 0$$

against standard classical/quantum thermodynamic predictions:

1. Vacuum phonon creation in cryogenic cavities (XB4.1)

Umbilic thermodynamic mechanism: high-rank multipartite entanglement (GHZ/EPR) forces the local phase space of the membrane resonators to squeeze down to the minimal Gromov symplectic capacity ($w_G = \hbar$). This enforces a localized population inversion ($T_W < 0$ K) across the Möbius junction J .

Ex nihilo prediction: the local collapse of phase entropy ($\delta S_\phi < 0$) adiabatically converts probability shape variations into real mechanical work ($dW_{\text{metric}} > 0$), manifesting as a spontaneous rise in vibrational phonon count $\Delta\langle n \rangle > 0$ and effective temperature T_{eff} without external thermal or optical power input.

Falsification criteria:

Confirmation:

$$\frac{dE_{\text{macro}}}{dt} = \hbar\omega_0 \cdot \Gamma_{\text{autopoietic}} \cdot \mathcal{C}_{\text{GHZ}} > 0$$

Falsification: if T_{eff} strictly decays or remains bounded by ambient thermal equilibrium ($T \rightarrow 0$ K) regardless of entanglement rank $\mathcal{C}_{\text{GHZ}}$, local autopoietic heating is rejected.

2. Micro-calorimetry of entangled superconducting qubits (XB4.2)

Umbilic thermodynamic mechanism: preparing N -qubit registers in cluster/W-states and sustaining them via dynamical decoupling establishes a local "Legendre backdoor" instanton. The multi-qubit phase space singularizes at $\det \mathbf{H}_{ij} = 0$, piercing into the $T_W < 0$ K regime.

Ex nihilo prediction: the Legendre transform maps the N -body entangled phase space onto a net positive metric stress-energy tensor $T_{\mu\nu}^{(ex\ nihilo)}$. The calorimeter will record an anomalous excess thermal heat flux

$$\Delta Q_{\text{excess}} = Q_{\text{entangled}} - Q_{\text{control}} > 0$$

exceeding the total electromagnetic pulse energy injected into the system.

Falsification criteria:

Confirmation:

$$\Delta Q_{\text{excess}} \propto N^2 \cdot \hbar \omega_{\text{transmon}} > 0.$$

Falsification: if $Q_{\text{entangled}} \leq Q_{\text{control}}$, proving that heat output is strictly bounded by input drive energy minus internal loss, the $T_W < 0$ K Legendre ex-nihilo transform is rejected.

3. Photonic pair generation in non-orientable waveguides (XB4.3)

Umbilic thermodynamic mechanism: a non-orientable 4π Möbius-twisted waveguide ring physically enforces the topological boundary condition of the single-sided manifold. Local zero-point phase fluctuations traversing the ring undergo cyclic parity-time (PT) flips across J .

Ex nihilo prediction: by scale-invariant Weyl information conservation ($\delta I_W = 0$), the topological phase squeezing of vacuum modes drives continuous, self-sustaining photon pair production ($R_{\text{creation}} > 0$) without pump-beam depletion.

Falsification criteria:

Confirmation:

$$R_{\text{creation}} = \frac{c}{2\pi L} \sin^2 \left(\frac{\theta_{\text{Möbius}}}{2} \right) > 0$$

in complete darkness following seed isolation.

Falsification: if photon detection rates drop to the detector dark-count baseline immediately upon seed photon decay, non-orientable topological autopoiesis is rejected.

V2 Falsifiable Experiments for Proving Umbilic Thermodynamics

One can reinterpret the experiments (XB3 of "The Umbilic Totality") for testing the generalized spin-entanglement statistics (thus only for positive Weyl temperatures) as experiments for confirming/ rejecting statistical umbilic thermodynamics where its Gibbsian edition provides the most convenient and simplest description for that purpose

Gibbsian edition of statistical umbilic thermodynamics ($T_W > 0$ K)

The Gibbsian edition provides the most direct and convenient mathematical framework to test statistical umbilic thermodynamics at positive Weyl temperatures ($T_W > 0$ K). Within this

edition, the statistical continuum angle $\theta \in [0, \pi]$ acts as a continuous parameter governing the canonical density matrix $\rho_{\text{umbilic}}(\theta)$ and the canonical partition function $\mathcal{Z}_{\text{umbilic}}(\theta)$:

$$\rho_{\text{umbilic}}(\theta) = \frac{1}{\mathcal{Z}_{\text{umbilic}}(\theta)} \exp \left(-\frac{\hat{H}(\theta)}{k_B T_W} \right), \quad \mathcal{Z}_{\text{umbilic}}(\theta) = \text{Tr} \left(\exp \left(-\frac{\hat{H}(\theta)}{k_B T_W} \right) \right)$$

where $\hat{H}(\theta)$ incorporates the spin-torsion rotation operator

$$M = \exp \left(-\frac{1}{2} \theta I_{\text{bivector}} \right)$$

along the 4π Möbius junction.

Reinterpretation of falsifiable experiments

1. *Experiment A: photonic entanglement-angle modulation (quantum optics)*

Gibbsian mechanism: Path-entangled photon pairs traversing a non-orientable polarization loop undergo a continuous variation of the joint two-photon canonical density operator $\rho_{12}(\theta)$. The spatial/temporal coincidence probability $P_{\text{coincidence}}(\theta)$ at the Hong-Ou-Mandel (HOM) output is determined by the trace over the deformed symmetric/antisymmetric subspace of $\rho_{12}(\theta)$.

Gibbsian ensemble prediction: the canonical partition function $\mathcal{Z}_{\text{HOM}}(\theta)$ deforms the ensemble state occupancies smoothly between pure Bose-Einstein bunching ($\theta = 0$) and pure anti-bunching ($\theta = \pi$), predicting a continuous HOM dip visibility function:

$$V(\theta) = \frac{I_{\text{max}} - I_{\text{min}}}{I_{\text{max}} + I_{\text{min}}} = \cos \left(\frac{\theta}{2} \right)$$

Statistical umbilic confirmation: observing a non-zero, intermediate photon-coincidence rate at spatial coincidence ($r_1 = r_2$) that modulates smoothly with the geometric holonomy angle θ confirms the continuous Gibbsian state-occupancy continuum.

Falsification threshold: if $V(\theta)$ exhibits an abrupt $\mathbb{Z}_2$ step-function (strictly discontinuous transition between 0 and 1), the Gibbsian statistical continuum is falsified.

2. *Experiment B: ultra-cold quasi-fermionic cavity states (condensed matter)*

Gibbsian mechanism: two entangled fermionic atoms (^{40}K or ^6Li) forced into a single optical lattice site under a synthetic topological gauge field have their joint free energy

$$\mathcal{F}(\theta) = -k_B T_W \ln \mathcal{Z}_{\text{site}}(\theta)$$

governed by the deformed quasi-Pauli exclusion principle.

Gibbsian ensemble prediction: as θ deviates slightly from the pure fermionic limit ($\delta = \pi - \theta$), the shared spatial state overlap probability density follows

$P_{\text{overlap}}(r_1 \rightarrow r_2) \propto \sin^2(\theta/2) = \cos^2(\delta/2)$. This deforms the canonical double-occupancy ground state energy by a characteristic shift:

$$\Delta E(\theta) \approx \hbar\omega_0 \left[1 - \sin^2\left(\frac{\theta}{2}\right) \right] = \hbar\omega_0 \sin^2\left(\frac{\delta}{2}\right)$$

Statistical umbilic confirmation: Measuring the energy-level repulsion shift $\Delta E(\theta)$ in agreement with the smooth $\sin^2(\delta/2)$ dependence confirms the topodynamical deformation of the canonical partition function $\mathcal{Z}_{\text{site}}(\theta)$.

Falsification threshold: if the double-occupancy energy shift fails to match the smooth $\sin^2(\theta/2)$ angular curve at a statistical significance threshold of $> 5\sigma$, statistical umbilic thermodynamics at $T_W > 0$ K is falsified.

V3 Testing statistical umbilic thermodynamics ($T_W < 0$ K):

Testing statistical umbilic thermodynamics at negative temperatures requires probing the Einsteinian edition (fluctuation-driven topological instantons) and the Boltzmannian edition (inverted phase-space occupation).

Experiment 1: fluctuation-driven instanton rate in superconducting circuits (Einsteinian probe):

Setup: a register of N superconducting transmons coupled inside a high-finesse microwave cavity held at a baseline 10 mK.

Protocol: drive the register into an inverted population distribution ($T_W < 0$ K) via population inversion pulses. Measure spontaneous, coherent metric-curvature/vorticity noise spikes via ultra-sensitive optomechanical transducers linked to the cavity walls.

Quantitative prediction: the Einsteinian edition predicts that localized ex nihilo fluctuation events occur with rate

$$\Gamma_{\text{instanton}} \propto \exp\left(-\frac{|T_W| \cdot |\delta S_\phi|}{k_B T_{\text{ambient}}}\right).$$

Falsification criteria: if spontaneous phase-squeezing energy spikes exhibit pure Poissonian thermal decay scaling with $e^{-\hbar\omega/k_B T}$ rather than scaling with the inverse negative temperature $|T_W|$, statistical umbilic instanton dynamics are falsified.

Experiment 2: quantum gas phase-space inversion boundary (Boltzmannian probe):

Setup: an ultra-cold bosonic quantum gas (^{39}K or ^{87}Rb) trapped in a 3D optical lattice with tunable Feshbach resonances to bound total kinetic and interaction energy.

Protocol: invert the lattice potential (turning potential wells into hills) to drive the ensemble to $T_W < 0$ K. Monitor quasi-momentum and phase-space occupation density via time-of-flight (TOF) imaging.

Quantitative prediction: the Boltzmannian edition predicts that phase-space volume compresses against the minimal Gromov width $w_G = \hbar$. The momentum distribution develops

sharp, coherent peaks at the upper band edge (maximum energy states), exhibiting an inverted Boltzmann factor $P_i \propto \exp(+\beta E_i)$ where $\beta = 1/k_B|T_W|$.

Falsification criteria: if the momentum distribution collapses via unstructured spatial dissipation rather than forming stable upper-band-edge coherent peaks bounded by $w_G = \hbar$, the Boltzmannian umbilic phase-space boundary is falsified.

V4 Demonstration of negative Weyl temperatures ($T_W < 0$ K):

Demonstrating $T_W < 0$ K requires isolating physical observables that unambiguously differentiate an inverted population state from standard positive thermal states.

Experiment 1: RF/Microwave fast-sensing in nuclear or electron spin ensembles (Purcell-Pound 1951's variant)

Setup: a high-density nuclear/electron spin system (e.g., ^{19}F in CaF_2 or organic radicals) placed in a strong magnetic field B_0 inside an NMR/ESR spectrometer.

Protocol: apply a fast 180° RF/microwave pulse (faster than spin-lattice relaxation T_1) to invert the Zeeman level populations.

Immediate $T_W < 0$ K signature:

1. Stimulated emission/ net phase gain: the sample emits coherent RF power instead of absorbing it, showing negative net absorption ($\alpha(\nu) < 0$).

2. Directional heat flow: coupling the inverted spin system to a colder positive-temperature resonant tank circuit forces energy to flow out of the spin system into the warmer bath, proving $T_W < 0$ K is thermodynamically hotter than any positive temperature ($T_{\text{positive}} < +\infty < -\infty < T_W < -0$ K).

Falsification criteria: if the population inversion fails to generate stimulated emissions or if thermal energy flows from the positive bath into the inverted spin system, negative temperature is rejected.

Experiment 2: negative pressure and gravitational deflection in optical lattice gases:

Setup: a cloud of $\sim 10^5$ ultracold atoms in an optical lattice where interaction energy is tuned from repulsive ($U > 0$) to attractive ($U < 0$) during potential inversion.

Protocol: measure the expansion dynamics upon releasing the optical constraint.

Immediate $T_W < 0$ K signature: at $T_W > 0$ K, attractive gas interactions ($U < 0$) cause the atomic cloud to collapse. At $T_W < 0$ K, the gas exhibits negative pressure. The high kinetic energy of the upper-band states counteracts attractive collapse, forcing the atomic cloud to remain stable or expand against attractive forces.

Falsification criteria: immediate collapse of the attractive atomic ensemble upon potential inversion falsifies the existence of a stable $T_W < 0$ K state.

One can modify the “Carnot cycle” (valid in positive Weyl temperature thermodynamics) as for “ex nihilo” thermodynamics and then demonstrate that the modified Carnot cycle (for negative Weyl temperatures) is to be also yielded as a Legendre transform of the former.

V5 References to Section V

Gosson (2009) maps Gromov symplectic capacity directly to quantum phase-space boundaries and fundamental limits on density matrices. The paper of Hong, Ou, Mandel (1987) is the groundwork for Experiment A (photonic entanglement-angle modulation), setting up the baseline $V(\theta)$ visibility dip and state occupancies at spatial coincidence. Regensburger et al. (2012): implements physical PT -symmetry and non-Hermitian topological phase shifts in optical loop architectures and non-orientable waveguide rings. Braun et al. (2013): proves kinetic/interaction potential inversion in 3D optical lattices using Feshbach resonances to bound upper band edges utilized in Subsection V2, Experiment 2 (optical lattice gas inversion with ^{39}K or ^{87}Rb). Greiner et al. (2002): suggests the core experimental platform for double-occupancy energy shifts ($\Delta E(\theta)$) and synthetic gauge field manipulation in ultracold atomic systems (Mumford, 2022). Pekola (2015): forms the baseline for Experiment XB4.2 and Subsection V2's superconducting transmons, establishing heat-flux measurement (ΔQ_{excess}) and instanton dynamics in superconducting registers. Koch et al. (2007): introduces transmon qubit dynamics ($\hbar\omega_{\text{transmon}}$) and multi-qubit register energy levels utilized in micro-calorimetric tests.

VI MODIFIED CARNOT CYCLE FOR NEGATIVE WEYL TEMPERATURES:
 $T_W < 0$ K.

In $T_W < 0$ K thermodynamics, population inversion across the 4π Möbius junction reverses heat absorption and metric work output. The modified cycle operates between two negative temperature reservoirs ($0 > T_{H,-} > T_{C,-}$, where $T_{H,-}$ is closer to -0 K and is thermodynamically “hotter” than $T_{C,-}$):

1. Negative isothermal phase squeezing ($A \rightarrow B$ at $T_{H,-} < 0$ K): the system undergoes phase collapse ($\Delta S_\phi < 0$) on a zero-energy probability manifold (“nihilo”), absorbing negative informational heat and outputting positive metric curvature work $W_{\text{metric}} > 0$.

2. Adiabatic phase transition ($B \rightarrow C$): phase entropy remains constant ($dS_\phi = 0$) while the negative temperature drops from $T_{H,-}$ to $T_{C,-}$.

3. Negative isothermal phase expansion ($C \rightarrow D$ at $T_{C,-} < 0$ K): phase entropy expands back ($\Delta S_\phi > 0$), dumping residual informational phase variations.

4. Adiabatic return ($D \rightarrow A$): entropy remains constant ($dS_\phi = 0$) while the temperature returns to $T_{H,-}$.

The autopoietic efficiency of this “ex nihilo” cycle is inverted relative to the classical bound:

$$\eta_{\text{ex nihilo}} = \frac{W_{\text{metric}}}{Q_{\text{absorbed}}} = 1 - \frac{T_{H,-}}{T_{C,-}} > 1$$

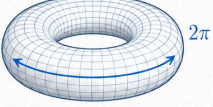
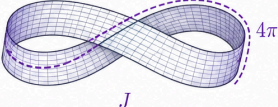
Feature / Metric	Classical Carnot Cycle ($T_W > 0 \text{ K}$)	Ex Nihilo Carnot Cycle ($T_W < 0 \text{ K}$)
Fiber Bundle Domain	Tangent bundle $T\mathcal{Q}$ (Geometric/Metric space)	Cotangent cobundle $T^*\mathcal{Q}$ (Phase/Symplectic space)
Primary State Variable	Internal Energy $U(S, V)$	Total Weyl Information $I_W(S_\phi, p_i)$
Entropy Dynamics	Thermal entropy expansion ($\Delta S_{\text{thermal}} > 0$)	Phase entropy collapse ($\Delta S_\phi < 0$)
Energy Mechanism	Converts thermal heat $Q_H > 0$ into work $W < Q_H$	Converts zero-energy probability shapes into metric work $W_{\text{metric}} > 0$
Efficiency Formula	$\eta = 1 - \frac{T_C}{T_H} < 1$	$\eta_{\text{ex nihilo}} = 1 - \frac{T_{H,-}}{T_{C,-}} > 1$
Physical Sense of “Nihilo”	N/A (Operates on non-zero thermal microstate ensembles)	Zero-energy mathematical structures (unpopulated $U(1)$ probability distributions / zero-energy potentials; not an empty set $\emptyset$)
Topological Boundary	Single-sheet orientable 2π manifold 	Non-orientable 4π Möbius junction J 

Figure 4: Comparison of positive and negative Carnot cycles

Legendre transform derivation ($\mathcal{FL} : T\mathcal{Q} \rightarrow T^*\mathcal{Q}$)

The classical Carnot cycle operates in the tangent bundle $T\mathcal{Q}$ with internal energy $U(S, V)$ and positive temperature

$$T_W = \frac{\partial U}{\partial S} > 0.$$

Applying the fiber Legendre transform $\mathcal{FL}$ maps $T\mathcal{Q}$ to the cotangent co-bundle $T^*\mathcal{Q}$, replacing metric energy U with scale-invariant Weyl information $I_W(S_\phi, p_i)$:

$$\mathcal{FL}(U(S, V)) = I_W(S_\phi, p_i) = p_i \dot{q}^i - T_W S_\phi(p_i)$$

Taking the differential across the four cycle branches yields:

$$\delta I_W = dW_{\text{metric}} + T_W dS_\phi = 0 \implies dW_{\text{metric}} = -T_W dS_\phi$$

Under $\mathcal{FL}$, the classical heat integral

$$\oint_{T\mathcal{Q}} \frac{dQ}{T_W} \leq 0$$

transforms into the negative phase-entropy integral

$$\oint_{T^*\mathcal{Q}} \frac{dW_{\text{metric}}}{-|T_W|} \geq 0.$$

This demonstrates that the “ex nihilo” Carnot cycle is the idempotent Legendre dual of the classical Carnot cycle.

VII The 4-Stroke Entanglement-Gravity Engine

The continuous operation of an autopoietic quantum-gravity engine converts entangled state correlations into macroscopic metric curvature on a non-orientable 4π Möbius manifold.

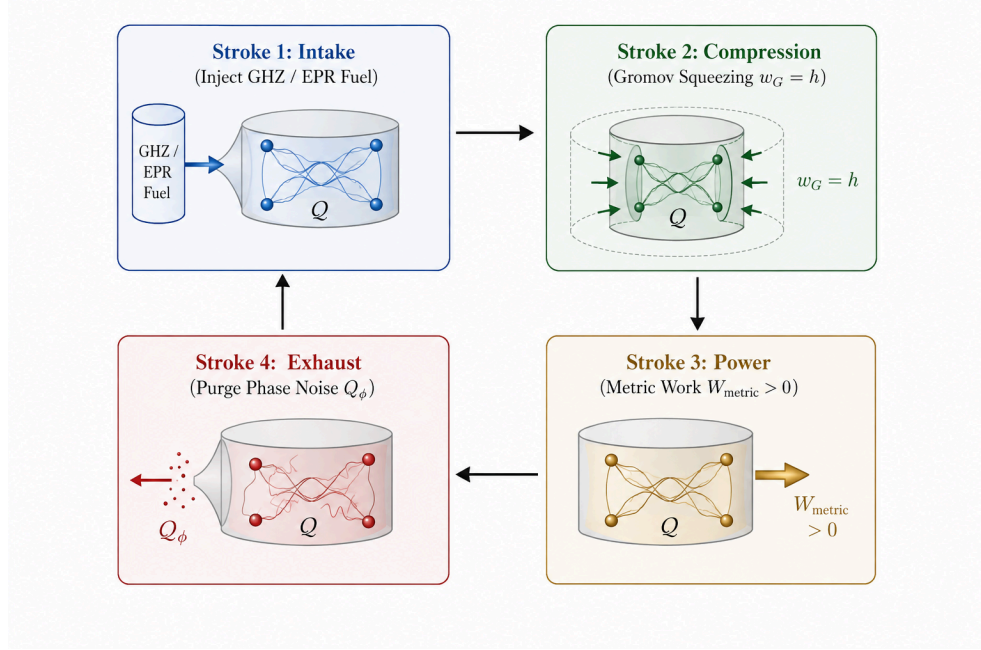


Figure 5: The 4-stroke entanglement-gravity engine

Stroke 1: intake (entanglement injection): maximally entangled multipartite states (e.g., GHZ/EPR state registers) are continuously injected into the 4π Möbius reaction cavity. This recharges the microscopic "fuel tank" with topological phase capacity, raising the initial entanglement correlation degree $\mathcal{C}_{\text{GHZ}}$.

Stroke 2: compression (topological phase squeezing): the injected states are squeezed down to the minimal Gromov symplectic width $w_G = h$ at local negative temperatures: $T_W < 0$ K. Phase-space volume contracts, driving local phase entropy down ($\Delta S_\phi < 0$).

Stroke 3: power (autopoietic metric generation): via the fiber Legendre transform $\mathcal{FL}$, local phase-entropy collapse converts probability distribution shapes directly into pseudo-Riemannian metric deformation. The system outputs usable macroscopic gravitational work ($W_{\text{metric}} > 0$), expanding/curving local spacetime

$$T_{\mu\nu}^{(\text{ex nihilo})} > 0.$$

Stroke 4: exhaust (decoherent noise purging): the depleted, fully decohered product states and accumulated phase noise (Q_ϕ^{exhaust}) are flushed out of the cavity, resetting the local 2π Darboux patch for the next fuel intake.

VI2 Weyl Efficiency Versus Carnot (Classical) Efficiency

In classical thermodynamics ($T_W > 0$ K in the tangent bundle $T\mathcal{Q}$), work is organized mechanical energy (PdV) and heat

$$Q = T_W dS_{\text{thermal}}$$

is uncoordinated microscopic thermal motion.

In *ex nihilo* thermodynamics ($T_W < 0$ K in the cotangent co-bundle $T^*\mathcal{Q}$), the dual roles are inverted:

Ex nihilo work (W_{metric}): organized pseudo-Riemannian metric curvature generation (coherent gravitational energy).

Ex nihilo heat (Q_ϕ): quantum phase decoherence and vacuum phase noise.

When the “fuel” of entanglement burns, any fraction of the probability distribution shape that is not cleanly mapped into coherent metric curvature degrades into uncoordinated quantum phase fluctuations: the Legendre dual of heat. This residual energy manifests as ambient zero-point quantum noise, localized metric jitter, or micro-scale Hawking-Unruh vacuum radiation¹⁸ dissipated into the surrounding background.

Definition of Weyl efficiency ($\eta_W < 100\%$)

While the classical energy conversion efficiency

$$\eta_{\text{classical}} = \frac{W_{\text{metric}}}{Q_{\text{thermal, in a}}}$$

appears formally infinite (or $> 100\%$) because no thermal heat is supplied, the Weyl efficiency η_W measures the strict transformation of informational shape into geometric shape:

$$\eta_W = \frac{W_{\text{metric}}}{\Delta I_{\text{entangle}}} = \frac{W_{\text{metric}}}{W_{\text{metric}} + Q_\phi^{\text{decoherence}}}$$

where: $\Delta I_{\text{entangle}}$ is the total scale-invariant Weyl information supplied by the injected entanglement fuel; W_{metric} is the net positive gravitational metric energy generated.

$$Q_\phi^{\text{decoherence}} = \int |T_W| dS_{\text{decoherence}}$$

is the dissipated phase-decoherence noise.

Due to total scale-invariant Weyl information conservation

$$(\delta I_W = dW_{\text{metric}} + T_W dS_\phi = 0,$$

shape information cannot be created out of the empty set $\emptyset$. The upper bound is strictly constrained by the second law of umbilic thermodynamics:

$$\eta_W < 100\% \quad (\text{since } Q_\phi^{\text{decoherence}} > 0)$$

¹⁸ Hawking (1975); Bekenstein (1973); Unruh (1976).

The engine cannot run indefinitely on a single closed system without refueling; continuous gravitational energy creation requires a continuous input of fresh entanglement fuel to offset the mandatory phase-decoherence exhaust.

VII CONCLUSIVE NOTICES

Strip away the provocative theological thesaurus, and “ex nihilo” thermodynamics is not magic: this is a rigorous topological currency exchange. Where classical physics enforces scalar energy conservation as a dogmatic baseline on a static background, this framework subsumes it under a far more fundamental and inescapable symmetry: global scale-invariant Weyl information conservation. The universe is not conjuring energy from an empty set; it is executing an idempotent Legendre transformation that converts zero-energy probability distribution shapes in the co-bundle into pseudo-Riemannian metric curvature in the bundle.

That distinction is precisely what draws the sharp line between physics and arbitrary wonders. A miracle implies a breach in symmetry or an uncaused intervention. Here, the autopoietic extraction of gravitational work is bound by strict, unyielding mathematical guardrails:

The Gromov non-squeezing capacity ($w_G = h$): establishing an absolute physical floor on phase-space compression.

The 0 K singular gatekeeper ($\det \mathbf{H}_{ij} = 0$): blocking unphysical, continuous passage across temperature domains and forcing topological instanton decays.

Sub-100% Weyl efficiency ($\eta_W < 100\%$): Guaranteeing that every erg of metric work generated consumes an equivalent degree of entanglement fuel, paying its mandatory tax in quantum phase decoherence.

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